

GENERAL EMBRYOLOGY

FOR HO STUDENTS

Terminologies

- **Oocyte** :The female germ or sex cells , produced in the *ovaries*.
- **Sperm or spermatozoon**, refers to the male germ cell produced in the testes.
- **Zygote**. the union of an oocyte and a sperm during fertilization.
- **Gestational period** – from the last menstrual period until birth

Terminologies cont...

- **Morula** (L. *morus*, mulberry). This solid mass of 12 to 16 blastomeres is formed by cleavage of a zygote.
- **Blastocyst**: After 2 to 3 days, the morula enters the uterus. Soon a fluid-filled cavity, the *blastocystic cavity*, develops inside it.
- This change converts the **morula into a blastocyst**

Terminologies cont...

- **Implantation.** The process during which the blastocyst attaches to the *endometrium*, and subsequently embeds in it.
- **Gastrula:** During gastrulation (transformation of a **blastocyst into a gastrula**), a three-layered or trilaminar embryonic disc forms (ectoderm, mesoderm, and endoderm).

Terminologies cont...

- **Neurula** (Gr. *neuron*, nerve). The early embryo during the third and fourth weeks when the **neural tube** is developing from the **neural plate**.
- It is the first appearance of the nervous system and the next stage after the gastrula.
- **Conceptus** – structures that develop from the zygote, **both embryonic and extraembryonic**.

Terminologies cont....

- **Embryology:** it is the study of embryos; the study of the origin and development of an individuals.
- **Teratology:** is the division of embryology and pathology that deals with abnormal development (birth defects).

Periods of human embryology

- **Medical point of view;** 1st, 2nd, 3rd trimester, each consist of three months period.
- **Embryologist's point of view:**
 - **Period of the egg-** from time of *fertilization to formation of blastocyst and implantation.*
 - **Period of the embryo-** *1st to 8th week*
 - **Period of the fetus-** *9th week to birth.*

Gametogenesis and cell division

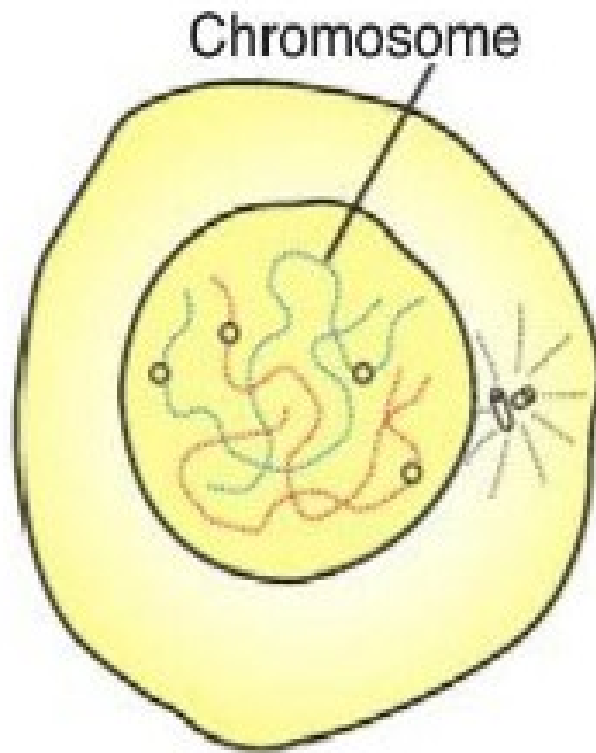
Mitosis: is the process whereby one cell divides, giving rise to two daughter cells that are genetically identical to the parent cell. Each daughter cell receives the complete complement of 46 chromosomes

- Before a cell enters mitosis, each chromosome replicates its deoxyribonucleic acid (DNA).
- During this replication phase, the chromosomes are extremely long, and they cannot be recognized with the light microscope.

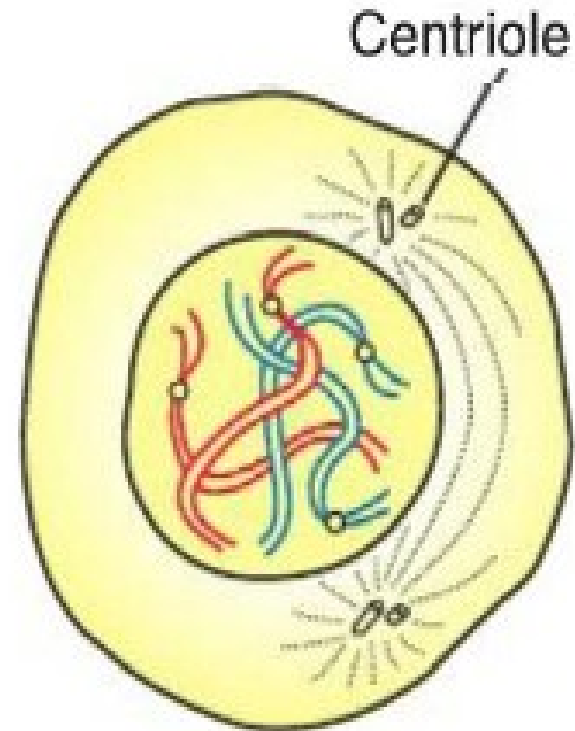
Stages of Mitosis

1. Prophase: the chromosomes begin to **coil, contract, and condense.**

- Each chromosome now consists of two parallel subunits, chromatids.
- Throughout prophase, the chromosomes continue to **condense, shorten, and thicken** but only at **prometaphase** do the chromatids become distinguishable.



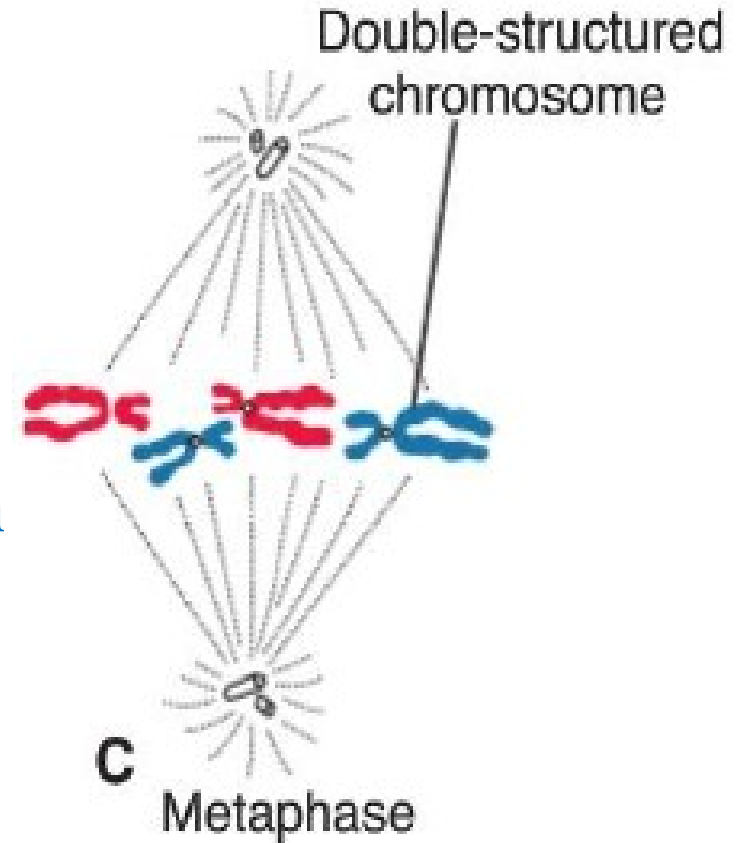
A
Prophase



B
Prometaphase

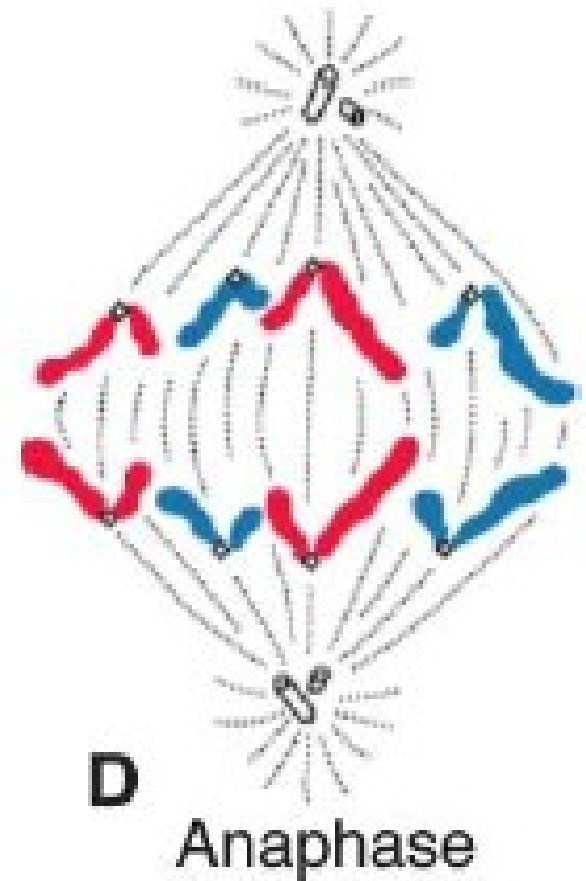
2. Metaphase

- During metaphase, the chromosomes line up in the **equatorial plane, and their doubled structure** is clearly visible.
- Each is attached by **microtubules extending from the centromere to the centriole**, forming the mitotic spindle



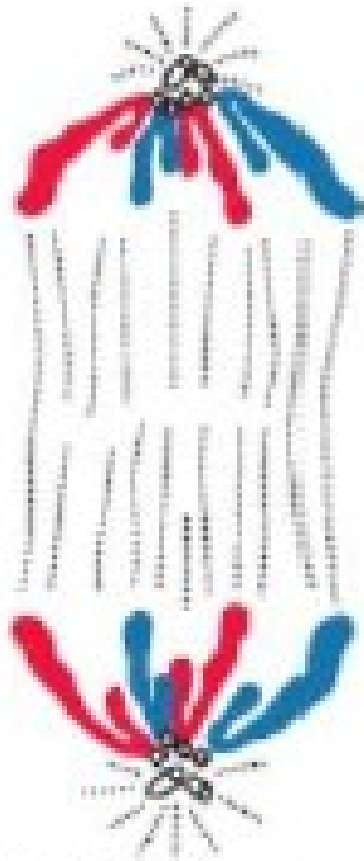
3. Anaphase

- The centromere of each chromosome divides, followed by **migration of chromatids to opposite poles** of the spindle.

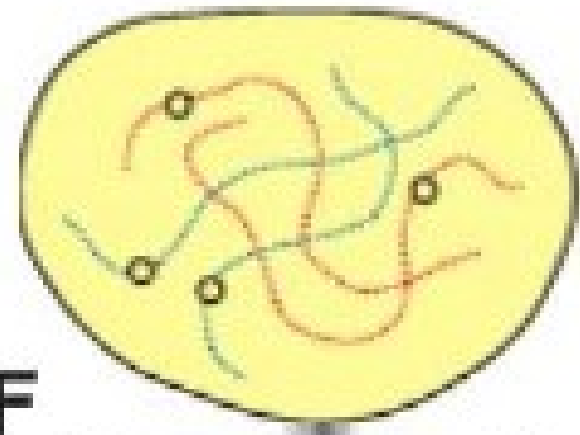
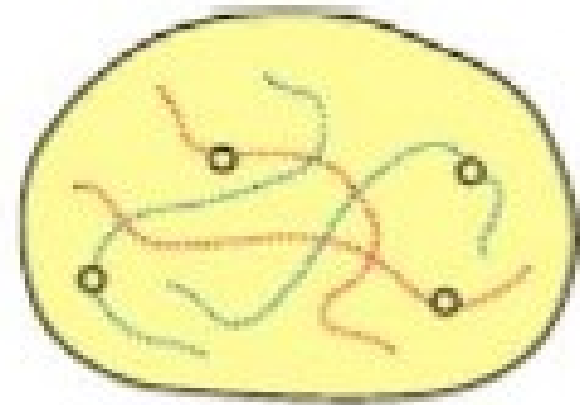


4. Telophase

- Finally, during telophase, **chromosomes uncoil and lengthen, the nuclear envelope reforms, and the cytoplasm divides.**
- Each daughter cell receives half of all doubled chromosome material and thus maintains the same number of chromosomes as the mother cell.



E
Telophase

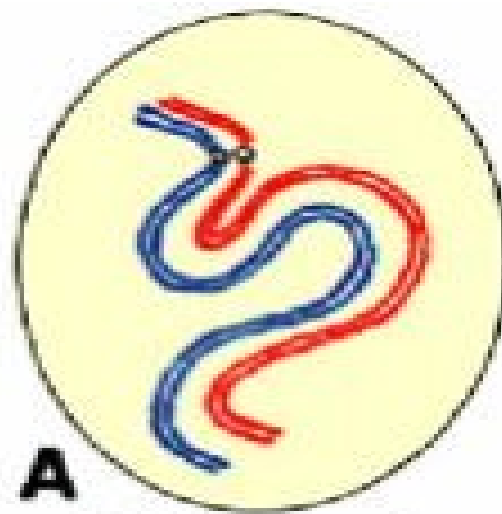


F
Daughter cells

Meiosis: is the cell division that takes place in the germ cells to generate male and female gametes.

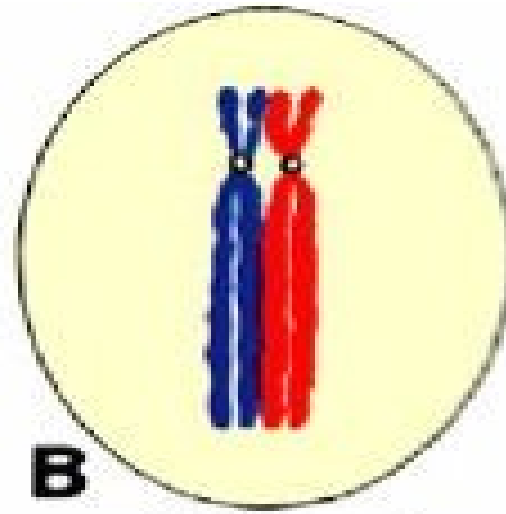
- Meiosis requires two cell divisions, meiosis I and meiosis II, to reduce the number of chromosomes to the haploid number of 23.
- As in mitosis, male and female germ cells (spermatocytes and primary oocytes) at the beginning of meiosis I replicate their DNA so that each of the 46 chromosomes is duplicated into sister chromatids.

- In contrast to mitosis, however, homologous chromosomes then align themselves in pairs
- Homologous pairs then separate into two daughter cells.
- Shortly thereafter, meiosis II separates sister chromatids.
- Each gamete then contains 23 chromosomes.



A

Pairing begins



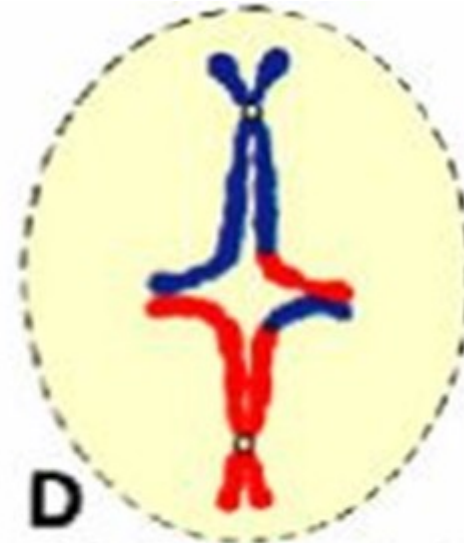
B

Pairing of chromosomes ✓



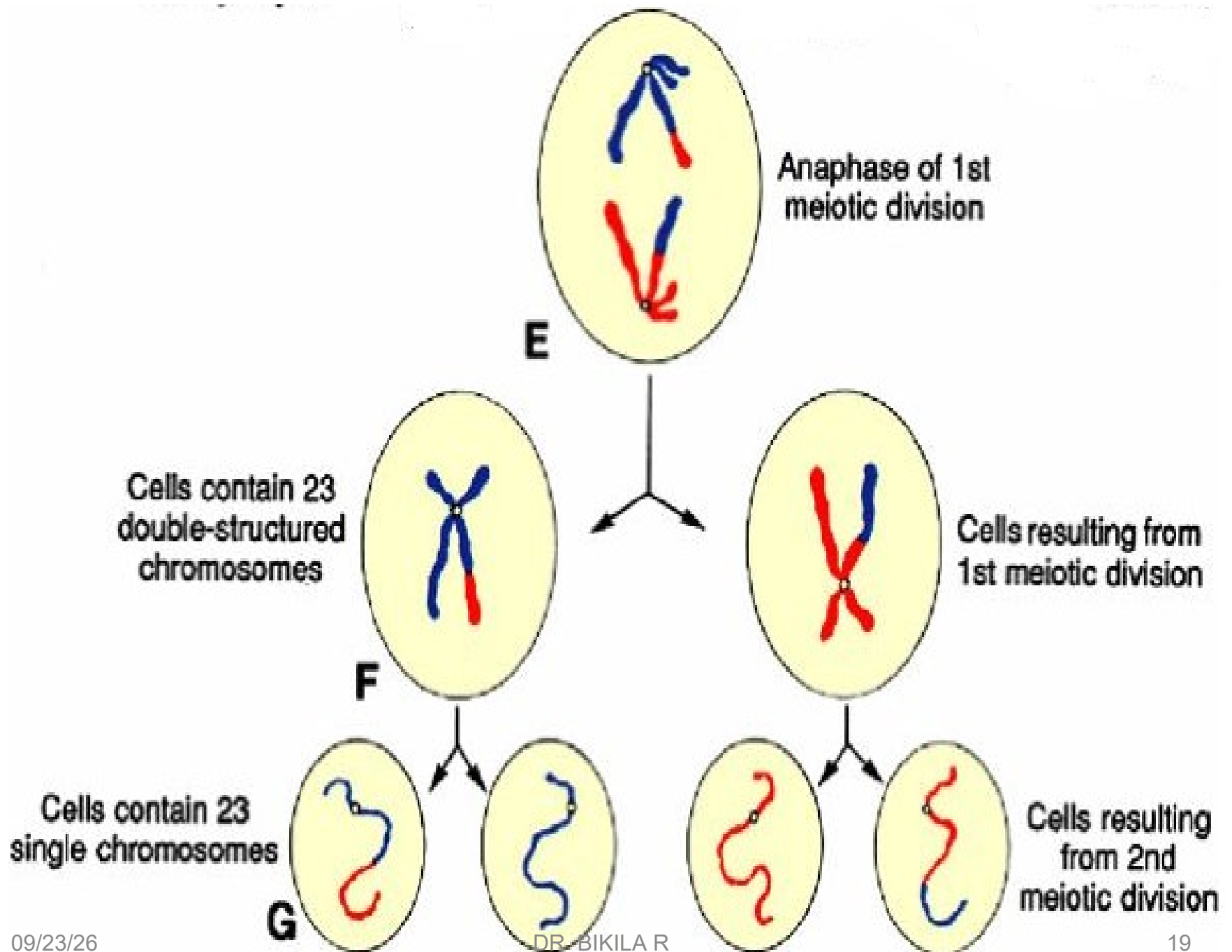
C

Chiasma formation



D

Pulling apart of double-structured chromosomes



Primordial germ cells (PGCs)

- Gametes are derived from **primordial germ cells (PGCs)** that are formed in the **epiblast** during the 2nd week and that move to the wall of the *yolk sac*.
- In 4th week PGCs begin to migrate from the yolk sac toward the **developing gonads**, arrive by the end of the 5th week.

GAMETOGENESIS

- Is conversion of germ cells into male and female gametes.
- *Timing of gametogenesis is different in males and females*

In males;

- PGCs remain dormant from the 6th week of embryonic development until puberty
- At puberty, seminiferous tubules mature and **PGCs** differentiate to **spermatogonia**.
- **Spermatogonia** undergo meiosis and mature into **spermatozoa**
- *Spermatozoa are produced continuously from puberty until death.*

GAMETOGENESIS cont....

In females;

- PGCs under go few **mitotic** division, and differentiate to **oogonia**
- In the *5th month of fetal development* all oogonia begin **meiosis** and they are called **primary oocyte**.
- **primary oocyte** remain **dormant** in stage of early meiosis **till puberty**.

GAMETOGENESIS cont....

- At puberty each month few ovarian follicles resume development in response to monthly **surge of pituitary gonadotropic hormones**.
- Usually only one primary oocyte mature in to **secondary oocyte** and is ovulated.
- The cycle continue till onset of **menopause**.

GAMETOGENESIS cont.....

- Gametogenesis in the male and female is called **spermatogenesis** and **oogenesis** respectively.
- The development of human being begins with **fertilization**, a process by which the **spermatozoon (male gamete)** and **oocyte (female gamete)** unite to give rise to a new organism, the *zygote*.

Oogenesis

- ***Prenatal maturation***

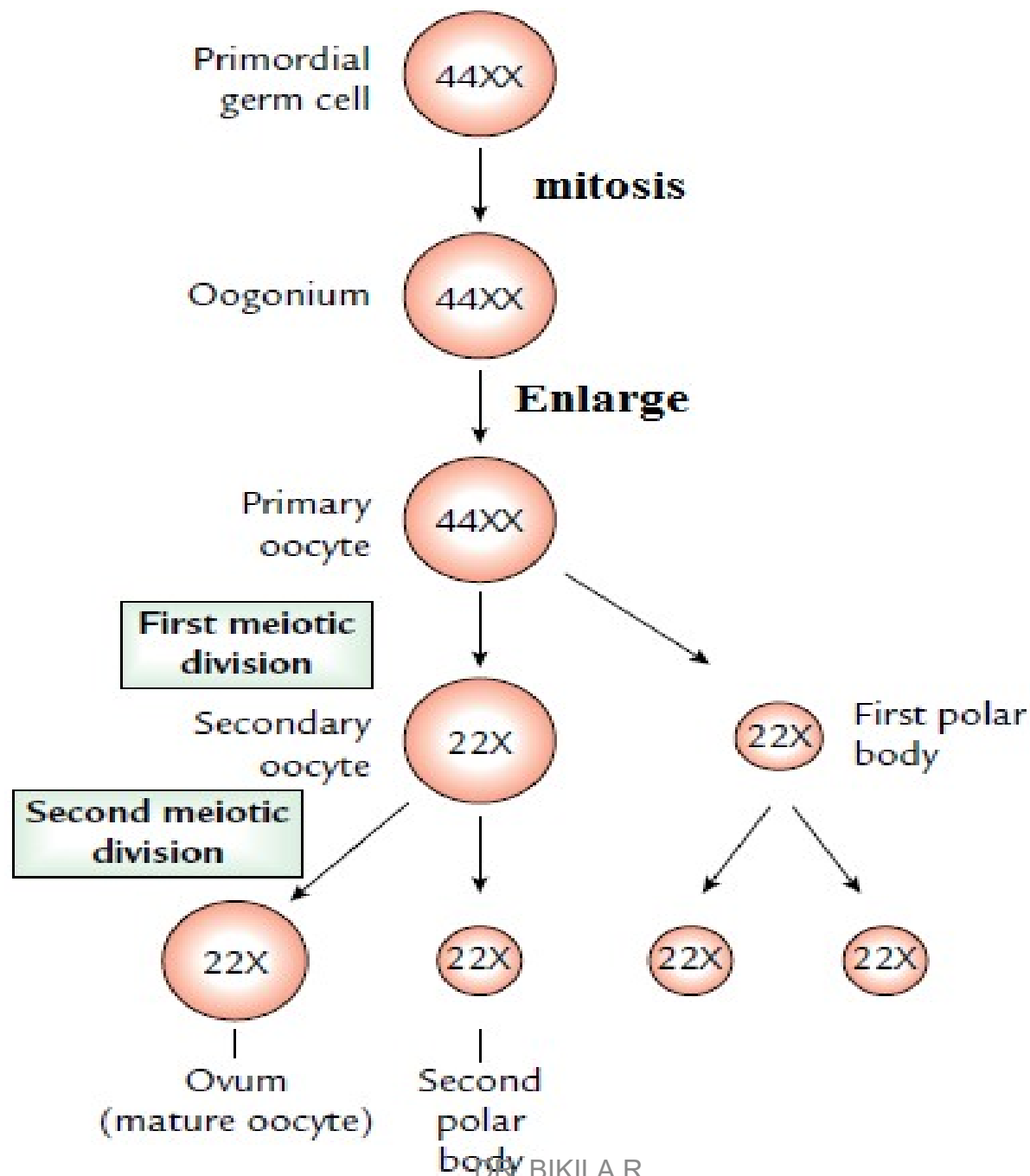
- During early fetal life, oogonia proliferate by mitosis. *Oogonia enlarge to form primary oocytes before birth;*
- Primary oocytes begin the first meiotic division before birth, but completion of prophase does not occur until adolescence.
- The follicular cells surrounding the primary oocyte are believed to secrete a substance, ***oocyte maturation inhibitor***, which keeps the meiotic process of the oocyte arrested.

Postnatal maturation

- Beginning during puberty, usually one follicle matures each month and ovulation occurs, except when oral contraceptives are used.
- *No primary oocytes form after birth in females*, in contrast to the continuous production of primary spermatocytes in males.
- The primary oocytes remain dormant in the ovarian follicles until puberty.

Postnatal maturation cont....

- As a follicle matures, the **primary oocyte** increases in size and, completes the first meiotic division to give rise to a secondary oocyte and the **first polar body**.
- Unlike the corresponding stage of spermatogenesis, however, the division of cytoplasm is unequal.
- The **secondary oocyte** receives almost all the cytoplasm and the first polar body receives very little.
- The polar body is a small, nonfunctional cell that soon degenerates



Postnatal maturation cont.....

- At ovulation, the nucleus of the secondary oocyte begins the second meiotic division, but progresses only to metaphase.
- If a sperm penetrates the secondary oocyte, the second meiotic division is completed.
- The other cell, the **second polar body**, soon degenerates.
- As soon as the polar body is extruded, maturation of the oocyte is complete.

Postnatal maturation cont.....

- *There are approximately two million primary oocytes in the ovaries of a newborn female, but most regress during childhood so that by adolescence no more than 40,000 remain.*
- Of these, only approximately 400 become secondary oocytes and are expelled at ovulation during the reproductive period.
- The number of oocytes that ovulate is greatly reduced in women who take oral contraceptives because the hormones in them prevent ovulation from occurring.

Polar Bodies

- During meiosis, one primary oocyte gives rise to four daughter cells, each with 22 plus 1 X chromosomes.
- However, only one of these develops into mature oocyte; the other three, the polar bodies, receive little cytoplasm and degenerate during subsequent development.
- Similarly, one primary spermatocyte gives rise to four daughter cells, two with 22 plus 1 X chromosomes and two with 22 plus 1 Y chromosomes.
- However, in contrast to oocyte formation, all four develop into mature gametes.

Spermatogenesis

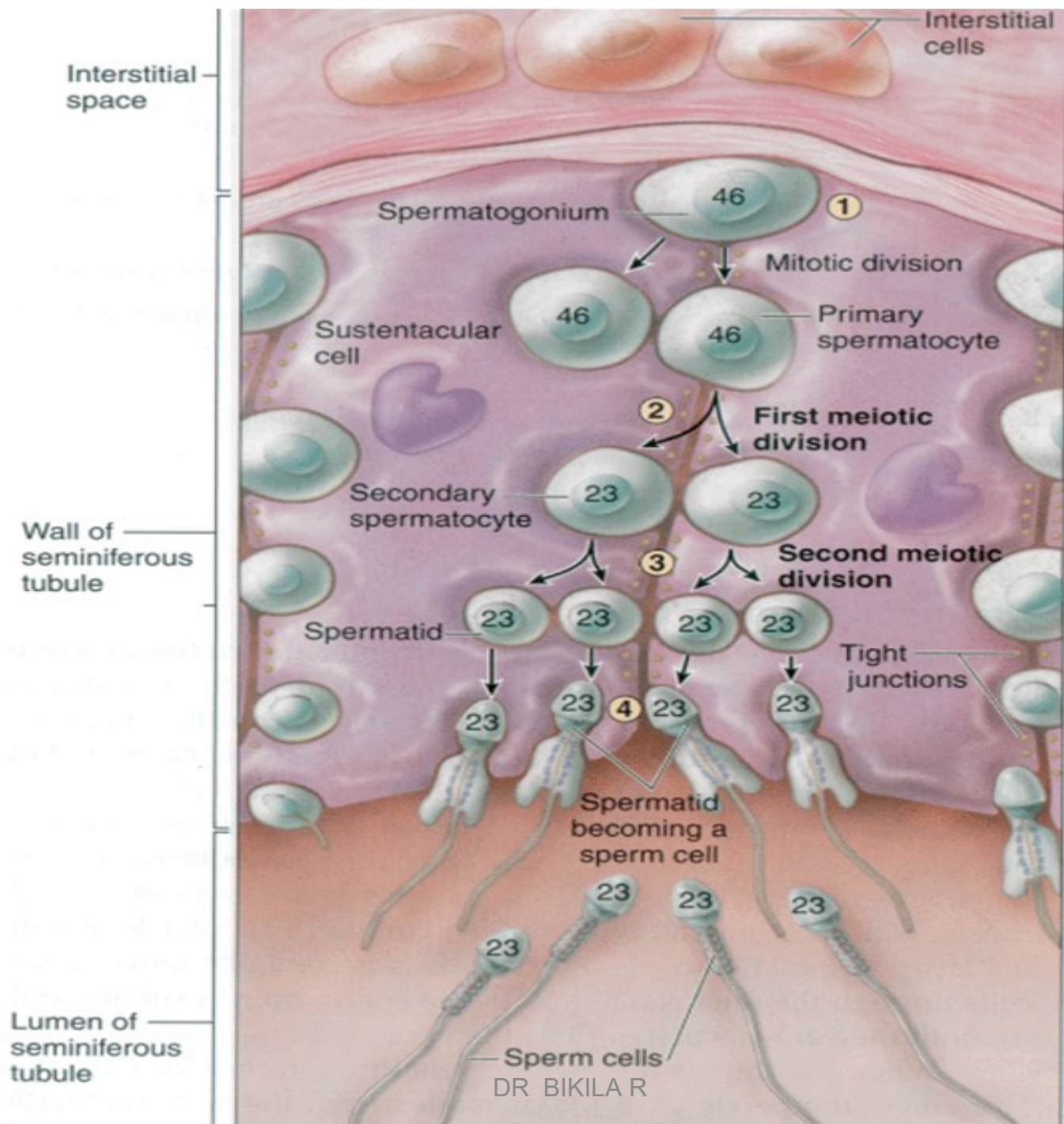
- Refers to the entire sequence of events by which primary germ cells (**spermatogonia**) are transformed into **spermatozoa**.
- It begins at puberty and continues to old age.
- The entire process of spermatogenesis, which includes spermiogenesis, takes approximately 2 months.

Spermatogenesis cont...

- **Spermatogonia**, which have been dormant in the seminiferous tubules of the testes since the fetal period, begin to increase in number at puberty.
- After several mitotic divisions, the spermatogonia grow and undergo changes.
- Spermatogonia are transformed into **primary spermatocytes**, the largest germ cells in the seminiferous tubules.

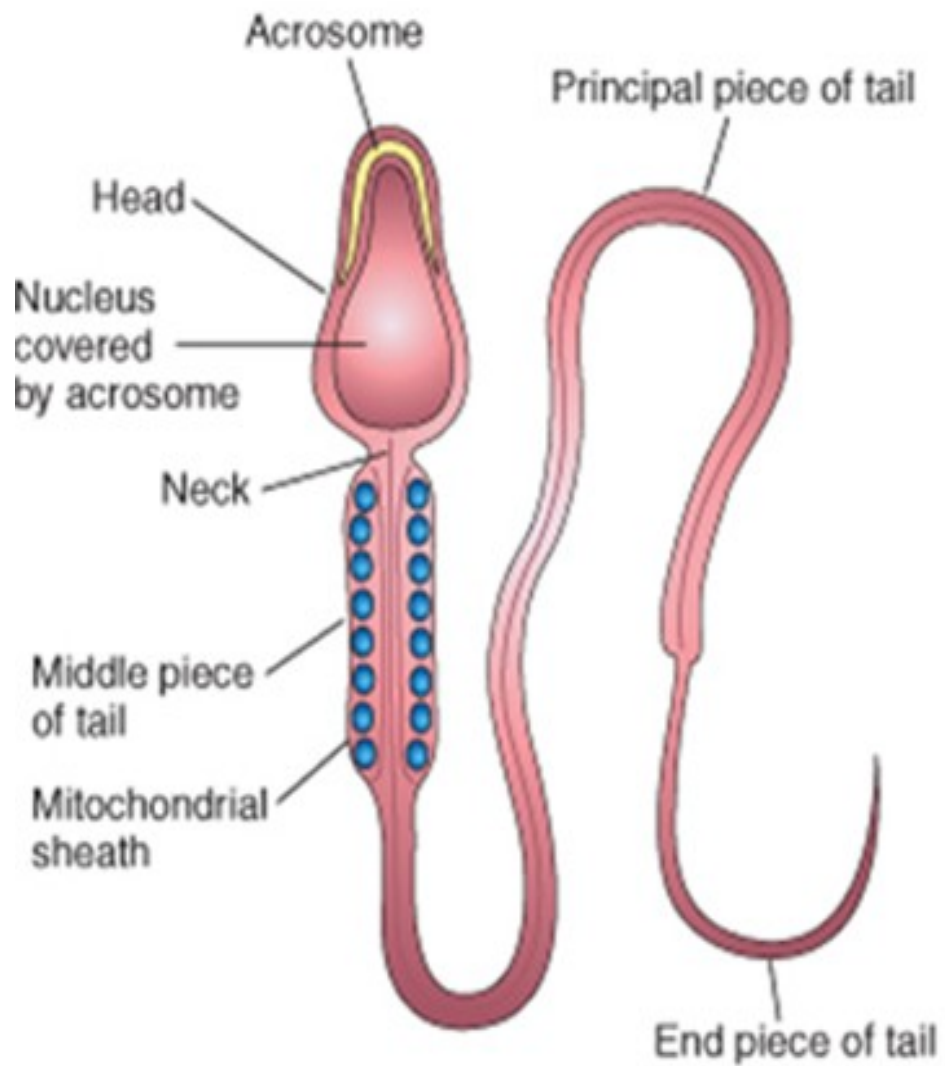
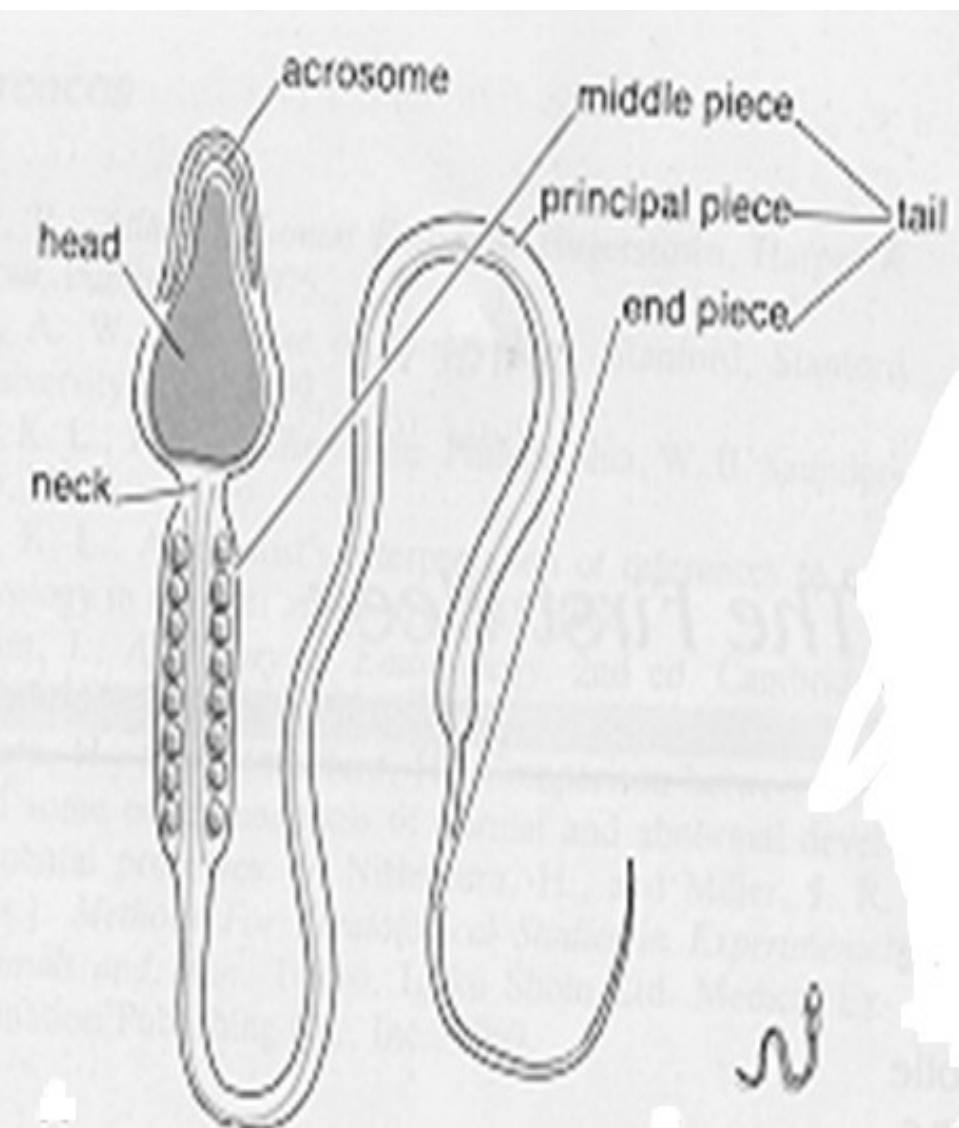
Spermatogenesis cont...

- Each primary spermatocyte subsequently undergoes a reduction division-the first meiotic division-to form two haploid **secondary spermatocytes**.
- Subsequently, the secondary spermatocytes undergo a second meiotic division to form four haploid spermatids.
- The spermatids are gradually transformed into four mature sperm by a process known as **spermiogenesis** .



The process of spermiogenesis include;

- a) Formation of the acrosome,** covers half of the nuclear surface and contains enzymes to assist in penetration during fertilization
- b) Condensation of the nucleus;** thickening and reduction of the nuclear size
- c) Formation of neck, middle piece, and tail;**
- d) Shedding of most of the cytoplasm:** elimination of all unnecessary cytoplasm.



- **Mature sperms** are free-swimming, actively motile cells consisting of **head, neck & tail**.
- The **head of the sperm** covered by the **acrosome**, a cap like saccular organelle containing several enzymes.
- These enzymes facilitate penetration of corona **radiata** and **zona pellucida** during fertilization.
- The **neck of the sperm** is the junction between the head and tail.

- The **tail of the sperm** consists of three segments: **middle piece, principal piece, and end piece.**
- The tail provides the motility of the sperm that assists its transport to the site of fertilization.
- The **middle piece of the tail** contains mitochondria, which provide the adenosine triphosphate necessary for activity.

Spermiogenesis cont...

- In a fertile male, approximately 100 to 200 million sperm are produced each day.
- Production of sperm (spermiogenesis & spermatogenesis) takes about 60 to 64 days
- An additional 12 to 21 days is needed for sperm to be transported into the epididymis, where they further mature and develop motility.

Hormonal factor that stimulate spermatogenesis

- Spermatogenesis is regulated by **luteinizing hormone (LH)** produced by the anterior pituitary.
- LH binds to receptors on **Leydig cells** and stimulates **testosterone production**, which in turn *binds to Sertoli cells to promote spermatogenesis*.
- **Follicle stimulating hormone (FSH)** is also essential because its binding to **Sertoli cells** stimulates **testicular fluid production** and synthesis of **intracellular androgen receptor proteins**

- **Sertoli cells** lining the seminiferous tubules **support and nurture the germ cells** and may be involved in the regulation of spermatogenesis.
- Sperms are transported passively from the seminiferous tubules to the *epididymis*, where they are **stored and become functionally mature**.

Hormonal factor...

- ***Growth hormone*** (as well as most of the other body hormones like **thyroid hormone**) is necessary for controlling background metabolic functions of the testes.
- Growth hormone specifically **promotes early division of the spermatogonia** themselves; in its absence, spermatogenesis is severely deficient or absent, thus causing infertility.

- **Normal semen analysis:**

Volume	>1.5 ml
Count	>20 million/ml
Motility	>50%
Morphology	>30%
WBC	<1 million/ml
PH	7-8

COMPARISON OF GAMETES

- **Oocyte**

- Massive cell when compared to spermatocyte
- Is immotile
- With respect to sex chromosome composition, only *one kind of normal secondary oocyte: 23, X*
- Surrounded by the zona pellucida and a layer of follicular cells, *the corona radiata*

- **Spermatozoa**

- Microscopic when compared to oocyte
- Highly motile
- With respect to sex chromosome composition there are *two kinds of normal sperm: 23, X and 23, Y.*
- No zona pellucida & corona radiata

First Week of Development

Ovulation to Implantation

Female internal genital organs

Ovaries

- ❖ oval, *almond-shaped gland, about 3cm long*
- ❖ Lies on the lateral wall of the pelvis minor before the first pregnancy
- ❖ The surface of the mature ovaries is irregular due to presence of **immature follicles, corpora lutea and scars of old corpora lutea**

Histology of the ovary

- The ovaries are surrounded externally by a fibrous **tunica albuginea**
- The tunica albuginea is in turn covered by a layer of cuboidal epithelium called **the germinal epithelium of the mesovarium**
- The ovary has an **outer cortex**, which houses the forming gametes, and an **inner medullary** region containing large blood vessels and nerves
- Embedded in the highly vascular connective tissue of the ovarian cortex have have **ovarian follicles**

Ovarian follicles

- Each follicle consists of an immature egg, called an oocyte encased in one or more layers of very different cells

Follicles at different stages of maturation are distinguished by their structure

1. Primordial follicle

- have one layer of squamous like follicle cells surrounding the oocyte

2. Primary follicle

- having two or more layers of cuboidal or columnar type granulosa cells enclosing the oocyte

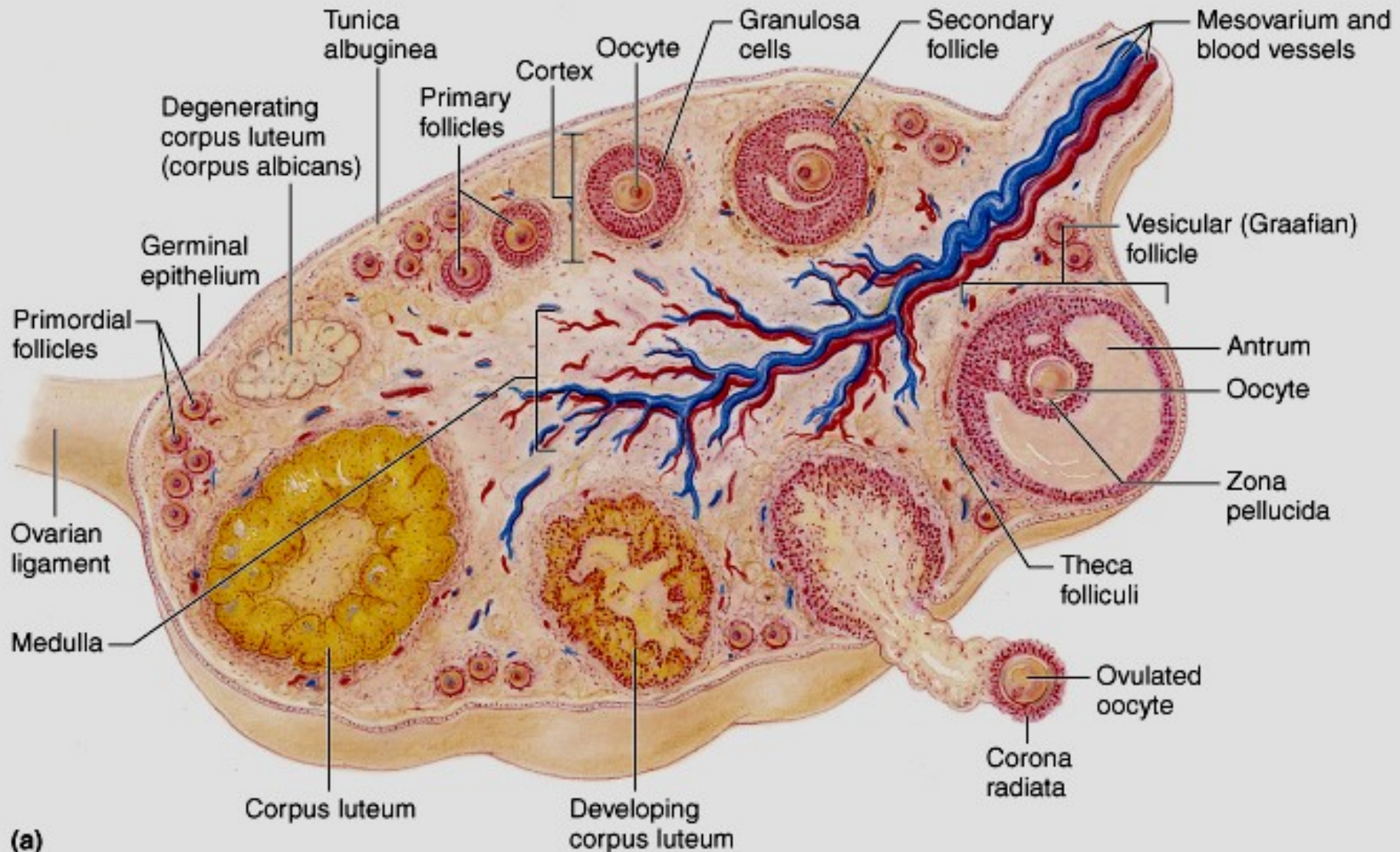
3. Secondary follicle

- have fluid filled spaces appearing between the **granulosa cells**

4. Mature follicle

- fluid filled spaces coalesce to form a fluid filled cavity called an antrum
- when it is called a **vesicular or Graafian follicle**, the follicle bulges from the surface of the ovary

- Each month in adult women, one of the ripening follicles ejects its oocyte from the ovary, an event called ovulation
- After ovulation, the ruptured follicle is transformed into a structure called the **corpus luteum** which eventually degenerates
- Most of these structures can be seen within the same ovary
- In older women, the surfaces of the ovaries are **scarred and pitted**, revealing that many oocytes have been released



(a)

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Uterus

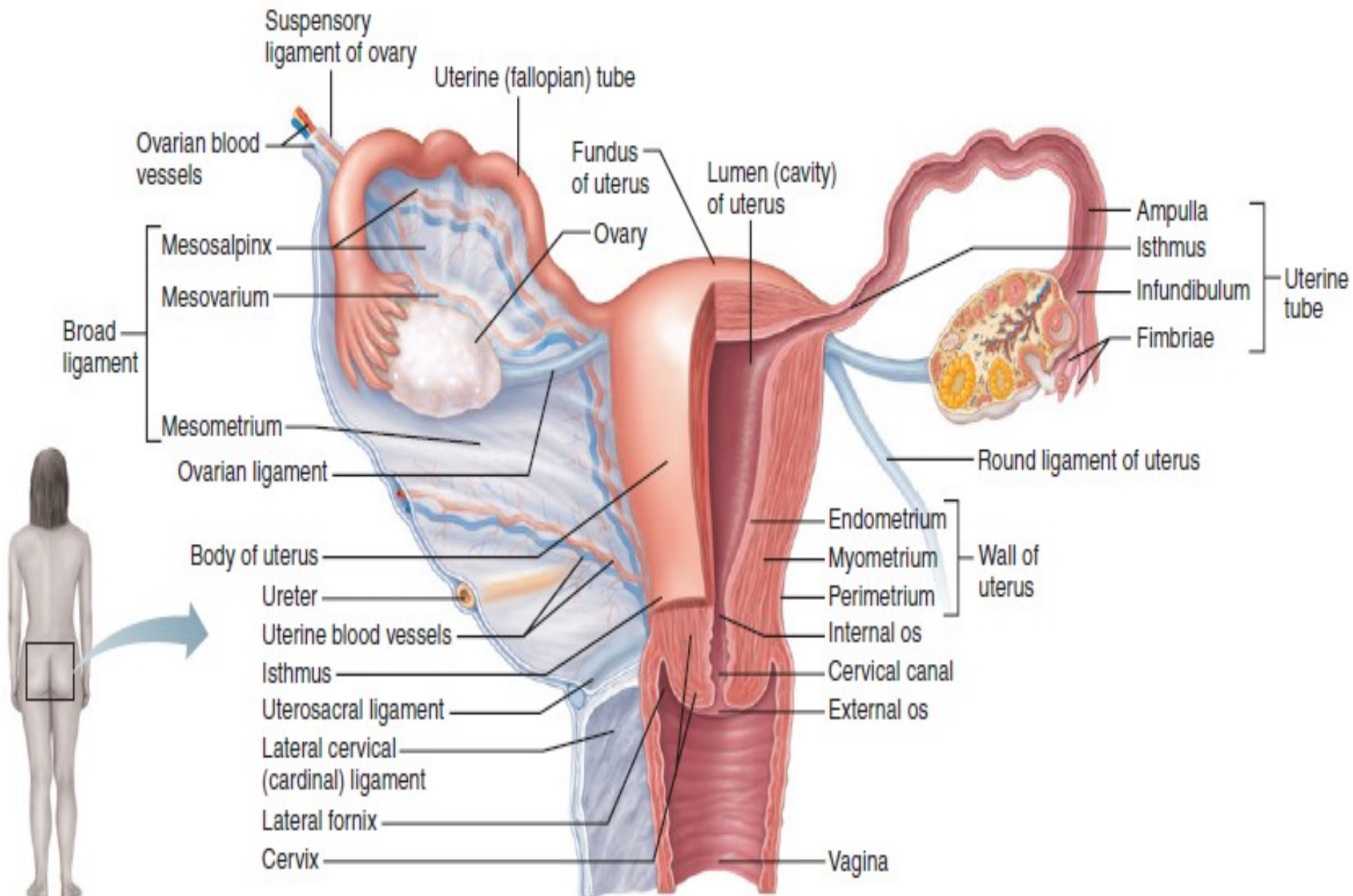
- ❖ Hollow, thick walled, pear-shaped muscular organ
- ❖ **Located between bladder and rectum**
- ❖ In a pre-menopausal woman who has never been pregnant, the uterus is about the size and shape of an **inverted pear**
 - 7-8cm long, 5-7cm wide, 3cm thick**
- ❖ It is larger in woman who have borne children
- ❖ Projects superoanteriorly over bladder

Parts

- **Fundus:** superior part of body
- **Body:** expanded *middle part*
- **Cervix:** cylindrical inferior third

Cervix

- ❖ Narrow neck, or outlet which projects into the vagina inferiorly
- ❖ Constriction between the cervix & body is *called internal os*
- ❖ The opening of the cervix into the vagina is called *external os*



- ❖ The vaginal portion of the cervix protrudes into the vagina, surrounded by the **anterior, posterior and lateral vaginal fornices**
- ❖ *The mucosa of the cervical canal contains **cervical glands** that secrete a mucus that fills the cervical canal and covers the external os*
- ❖ *The mucus block the spread of bacteria from the vagina into the uterus*
- ❖ *Cervical mucus also blocks the entry of sperm, except at mid cycle, when it becomes less viscous and allows sperm to pass through*

The Uterine Wall

- Made up of 3 layers

Perimetrium: outermost serous layer composed of visceral peritoneum

Myometrium: bulky middle layer, composed of interlacing bundles of smooth muscle

- contracts during childbirth to expel the baby

Endometrium: simple columnar epithelium underlined by a thick lamina propria of highly cellular connective tissue

- If fertilization occurs, the young embryo implants into the endometrium and resides there for the rest of development
- The endometrium has two chief strata

❑ **stratum functionalis (functional layer)**

- undergoes cyclic changes in response to blood levels of ovarian hormones and is *shed during menstruation (28 day cycle)*

❑ **stratum basale**

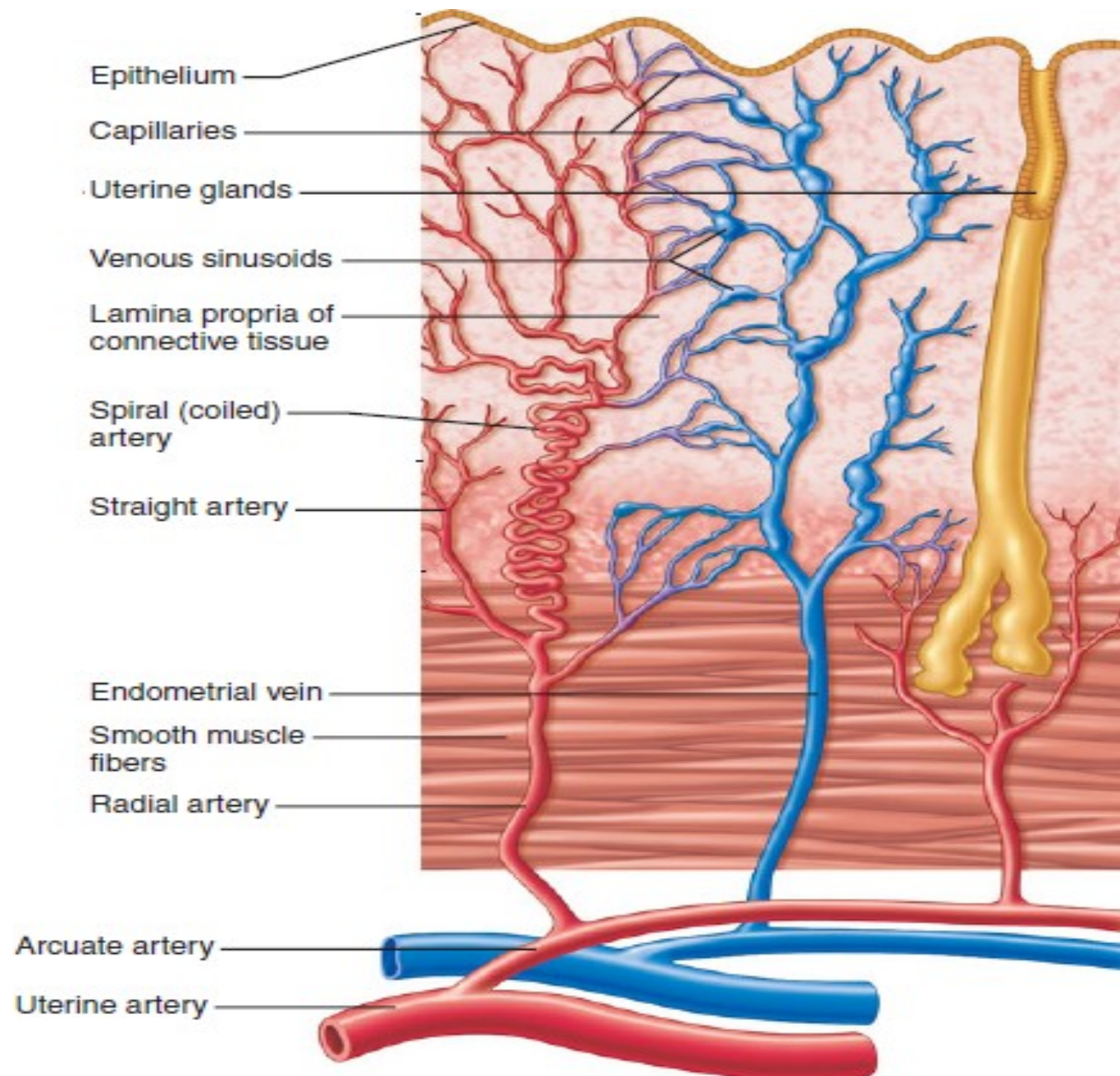
- thinner deeper layer which *forms a new functionalis layer after menstruation ends*
- *unresponsive to ovarian hormones*

Blood supply

- arterial supply: **uterine artery (main) and ovarian artery**
- venous drainage: **uterine vein**

uterine arteries

- arise from the **internal iliac** in the pelvis,
- Break up into several **arcuate arteries** within the **myometrium**
- arcuate arteries send **radial** branches into the endometrium
- Radial artery give off **straight arteries** to the stratum basalis **and spiral arteries** to the stratum functionalis
- **spiral arteries** undergo repeated degeneration and regeneration

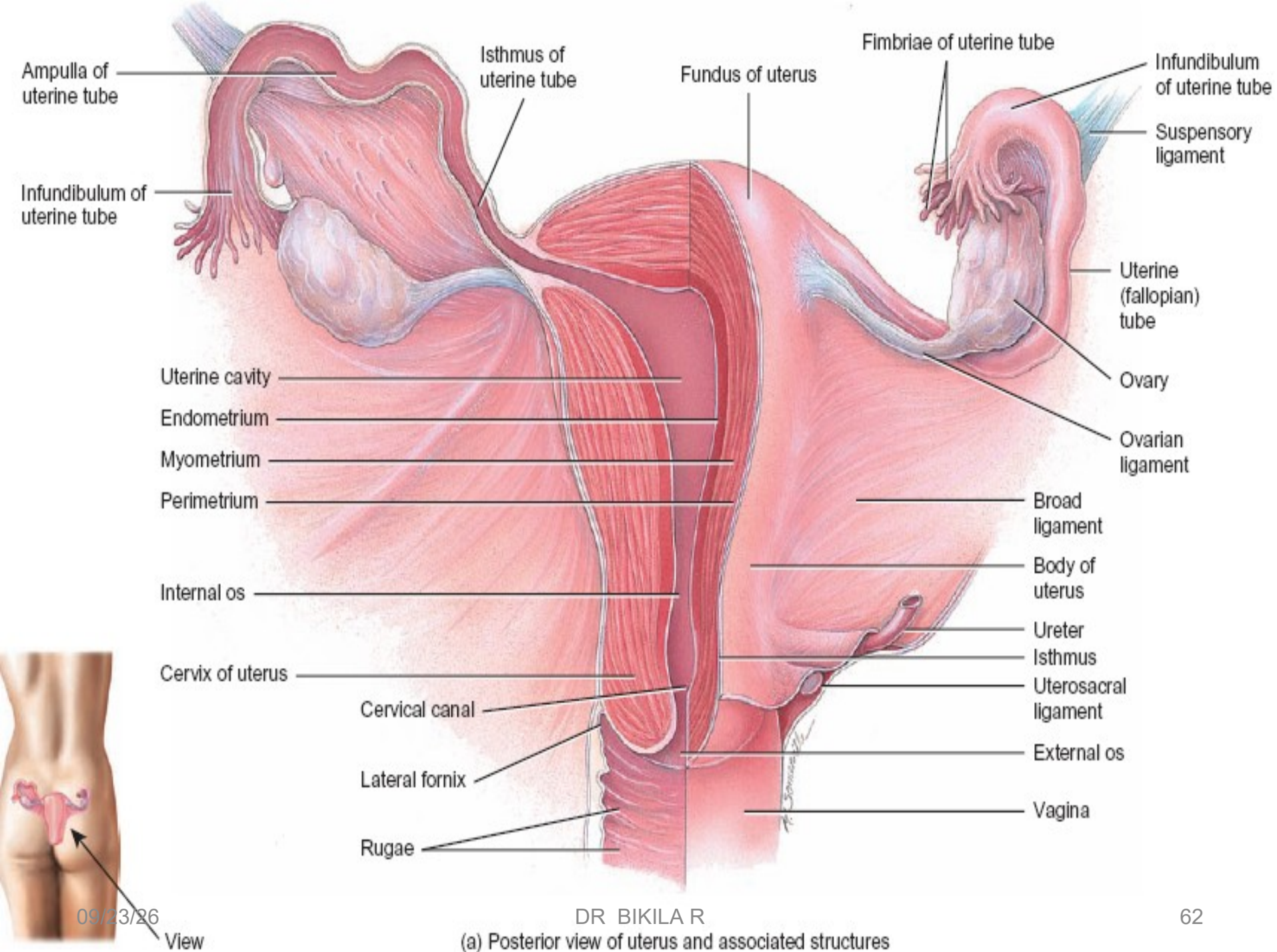


Uterine tubes

- 10-12cm long, extends medially from the ovary to uterus
- carry oocytes from ovaries and sperms from uterus to site of fertilization (ampulla)
- Opens at its proximal end to horn of uterus and distal end into peritoneal cavity

Parts:

- **Infundibulum:** funnel shaped
- **Fimbriae:** finger like processes which spread over the ovary
- **Ampulla:** middle wide part, site of fertilization
- **Isthmus:** narrow part which enters uterus



Wall

Histologically, the uterine tubes are composed of three layers:

- The **mucosa** consists of epithelium and lamina propria (areolar connective tissue). The epithelium contains ciliated simple columnar cells, which function as a cilia.
- The **muscularis**, is composed of an inner, thick, circular ring of smooth muscle.
- **Peristaltic contractions of the muscularis and the ciliary action of the mucosa** help move the oocyte or fertilized ovum toward the uterus.
- The outer layer of the uterine tubes is a serous membrane, **the serosa**.

FEMALE REPRODUCTIVE CYCLES

- At puberty, the female begins to undergo regular monthly cycles, involving activities of the **hypothalamus** of the brain, **pituitary gland** , **ovaries**, **uterus**, **uterine tubes**, **vagina** and **mammary glands**
- It prepares the reproductive system for pregnancy.
- These **sexual cycles** are controlled by the hypothalamic **Gonadotropin-releasing hormone (GnRH)**.
- **GnRH** acts on anterior pituitary gland, to secrete **gonadotropins (FSH & LH)**.
- **FSH** and **LH** stimulate and control cyclic changes in the ovary

FEMALE REPRODUCTIVE CYCLES cont...

- *Follicle-stimulating hormone (FSH)* stimulates the **development of ovarian follicles** and the **production of estrogen** by the follicular cells.
- *Luteinizing hormone (LH)* serves as the "**trigger**" for **ovulation** (release of secondary oocyte) and stimulates the follicular cells and corpus luteum to **produce progesterone**.

Ovarian cycle

○ *FSH and LH produce cyclic changes in the ovaries-the ovarian cycle, which include:*

- Development of follicles
- Ovulation, and
- Corpus luteum formation.

1. Follicular Development

○ Under normal conditions, only one of follicle reaches full maturity, and only one oocyte is discharged; the others degenerate and become atretic

- **Granulosa and thecal** cells produce **estrogens** that;
 - a. Cause the uterine endometrium to enter the follicular or **proliferative phase**;
 - b. Cause thinning of the cervical mucus to allow passage of sperm; and
 - c. Stimulate the pituitary gland to secrete LH.

Follicular Development...

- At mid-cycle, there is an **LH surge** that
 - a. Elevates concentrations of maturation-promoting factor, causing oocytes to complete meiosis I and initiate meiosis II;
 - b. Stimulates production of **progesterone** by follicular stromal cells (**luteinization**);
 - c. Causes follicular rupture and ovulation.

Follicular Development...

- In summary development of an ovarian follicle is characterized by:
 - Growth and differentiation of primary oocyte
 - Proliferation of follicular cells
 - Formation of zona pellucida
 - Development of the theca folliculi

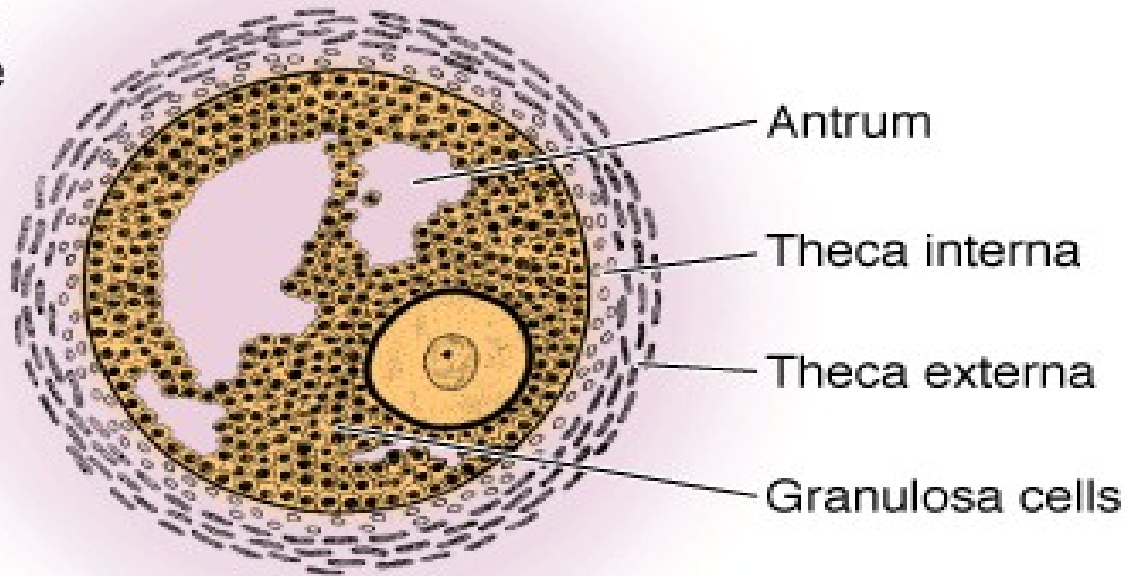
2. Ovulation

- Around midcycle, **the ovarian follicle**, *under the influence of FSH and LH*, undergoes a sudden *growth*, producing a cystic swelling or bulge on the surface of the ovary.
- The high concentration of **LH** increases **collagenase activity**, resulting in digestion of collagen fibers surrounding the follicle.
- **Prostaglandin** levels also increase in response to the **LH surge** and cause local **muscular contractions** in the ovarian wall.

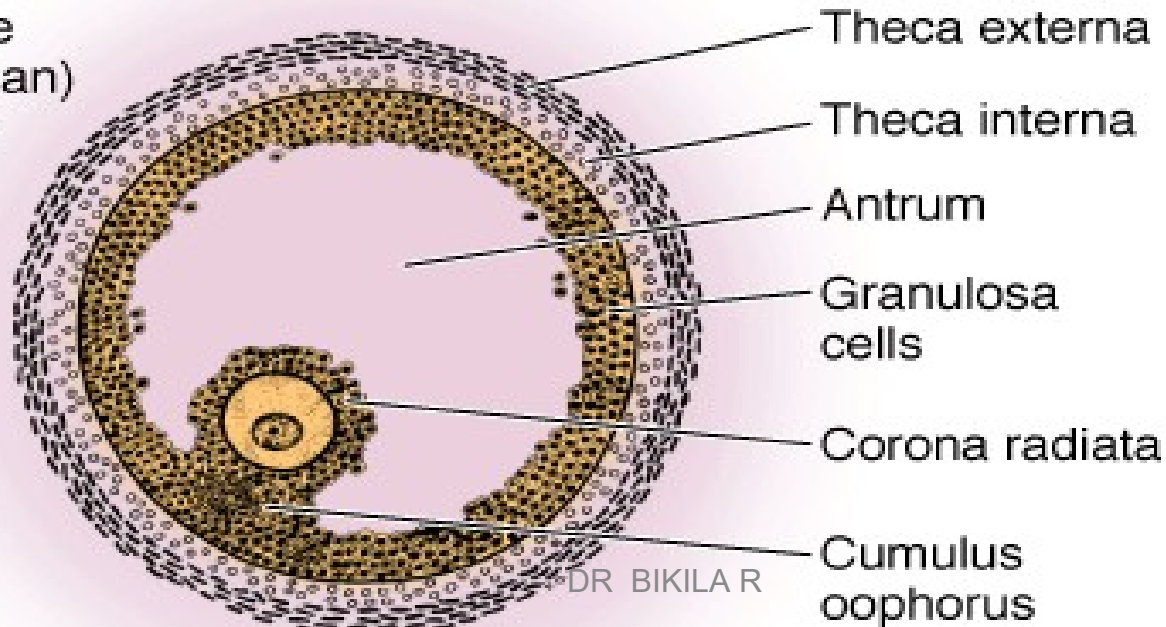
Ovulation cont...

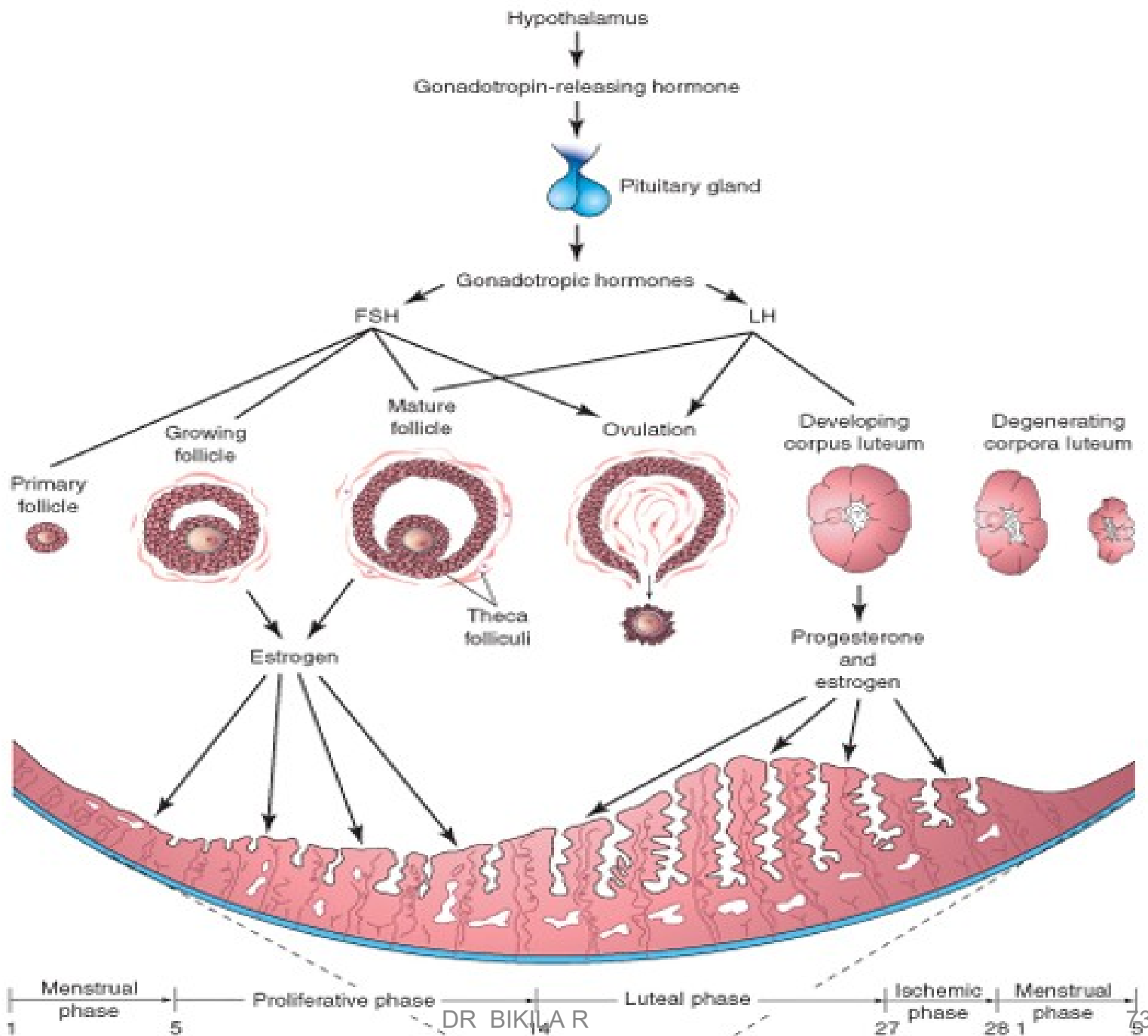
- Those contractions extrude the oocyte, which together **with its surrounding granulosa cells** from the region of the **cumulus oophorus**.
- Some of the *cumulus oophorus cells* then *rearrange* themselves around the **zona pellucida** to form the **corona radiata**.
- So expelled secondary oocyte is surrounded by the **zona pellucida** and one or more layers of follicular cells called **corona radiata**

Antral follicle



Mature (graafian) follicle





Ovulation cont...

Sign and symptom of ovulation

- ✓ Lower abdominal pain (Mittelschmerz)
- ✓ Slight increase in morning basal temperature
- ✓ Cervical mucus change (slippery)
- ✓ Slight bleeding to peritoneal cavity

Anovulation

Some women do not ovulate (cessation of ovulation-anovulation) because of an inadequate release of gonadotropins.

Ovulation cont...

- In some of these women, ovulation can be induced by the administration of **gonadotropins or an ovulatory agent such as clomiphene citrate.**
- This drug stimulates the release of pituitary **gonadotropins (FSH and LH)**, resulting in maturation of several ovarian follicles and multiple ovulations.
- The incidence of multiple pregnancy increases as much as **tenfold** when ovulation is induced

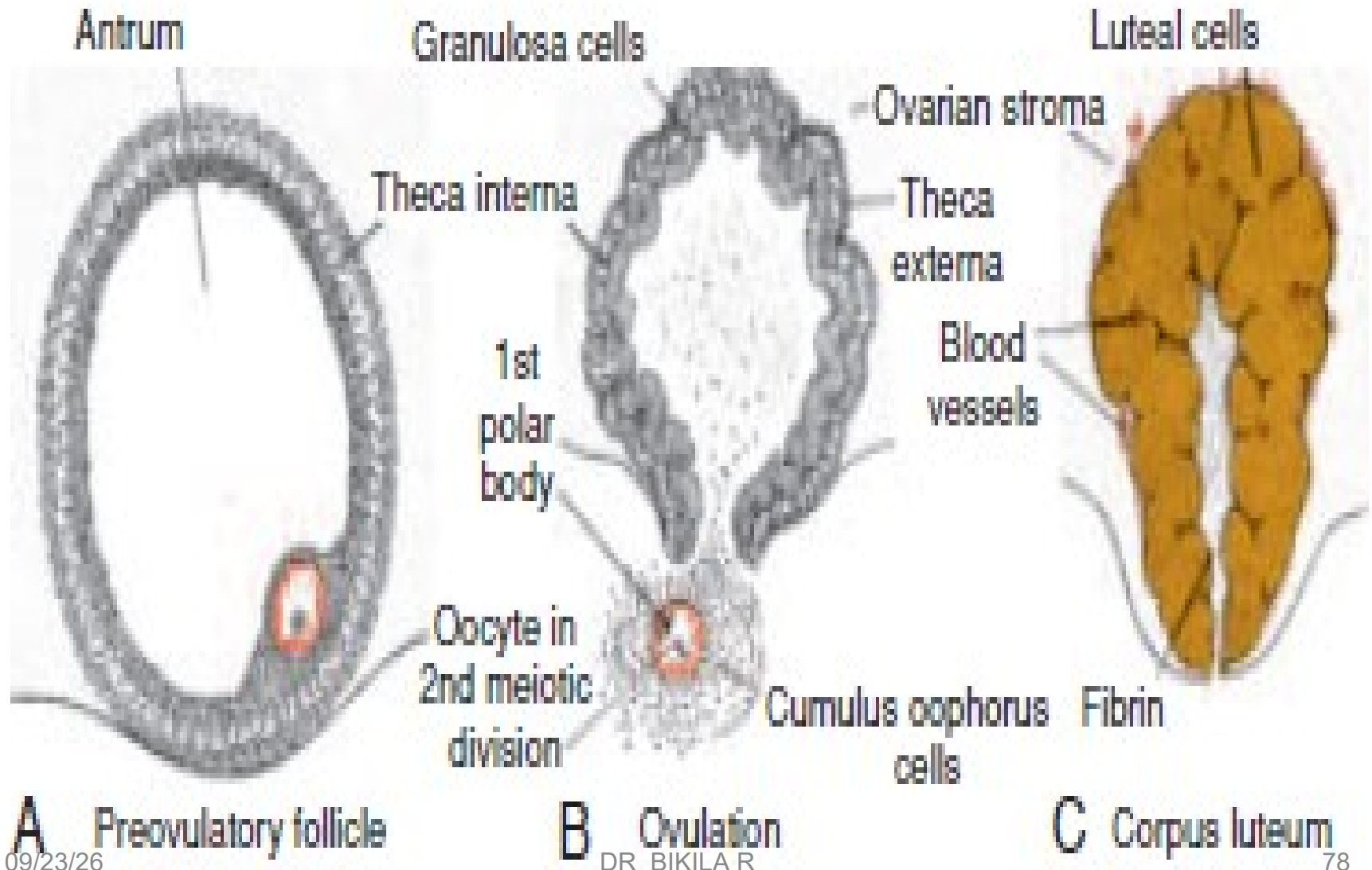
3. Formation of corpus luteum

- After ovulation, **granulosa cells** remaining in the wall of the ruptured follicle, together with cells from the **theca interna**, are vascularized by surrounding vessels.
- Under the influence of **LH**, these cells develop a ***yellowish pigment*** and change into a glandular structure, **lutean cells**(*form the corpus luteum and secrete progesterone.*)

Formation of corpus luteum cont...

- If the oocyte is fertilized, *degeneration* of the corpus luteum is prevented by ***human chorionic gonadotropin***, a hormone secreted by the **syncytiotrophoblast** of the blastocyst.
- The corpus luteum of pregnancy remains functionally active throughout the **first 20 weeks of pregnancy** till placenta takes the responsibility of hormone production to maintain the pregnancy.
- If the oocyte is not fertilized, the corpus luteum involutes and degenerates **10 to 12 days** after ovulation.

Ovarian cycle



Corpus Albicans_

- If fertilization does not occur, the corpus luteum reaches maximum development approximately 9 days after ovulation.
- Subsequently, the corpus luteum shrinks because of degeneration of lutean cells and forms a mass of fibrotic scar tissue, the **corpus albicans**.
- Simultaneously, progesterone production decreases, precipitating menstrual bleeding.

Uterine (endometrial) cycle

- The hormones produced by the ovarian follicles and corpus luteum (**estrogen and progesterone**) produce cyclic changes in the endometrium
- The average menstrual cycle is 28 days.
- Menstrual cycles normally vary in length by several days.
- In 90% of women, the length of the cycles ranges **between 23 and 35 days**

Uterine(endometrial) cycle cont...

- **Uterine cycle consist of;**
 - Menstrual phase
 - Proliferative phase
 - Secretory phase
- **Menstrual phase**
 - The functional layer of the uterine wall is sloughed off and discarded with the menstrual flow-menses (monthly bleeding), which usually lasts 4 to 5 days
 - It is the result of progesterone withdrawal (*corpus luteum degenerates if pregnancy doesn't occur and hormone level falls*).

Uterine(endometrial) cycle cont.....

- **Proliferative Phase**

- Coincides with growth of ovarian follicles
- On about day 5, **theca and follicular cells** of growing follicle secrete **estrogen**.
- This hormone cause endometrium to proliferate and undergo remodeling.
- This phase begins on day 5 and completed on day 14.
- After ovulation, **estrogen** and **progesterone** is produced by **corpus luteum**

Uterine(endometrial) cycle cont.....

- **Luteal(secretory) Phase**
 - Coincides with the formation, functioning, and growth of the corpus luteum
 - **Progesterone** causes endometrium to enter **secretory phase** (endometrium thickened, glands convoluted and vasculature increases)
 - This phase lasts about 13 days

Uterine(endometrial) cycle cont.....

– If fertilization does not occur:

- The corpus luteum degenerates.
- Estrogen and progesterone levels fall and the secretory endometrium enters an **ischemic phase**.
- Menstruation occurs.

Ischemic Phase

- The ischemic phase occurs when the oocyte is not fertilized.
- *Ischemia* (reduced blood supply) occurs as the spiral arteries constricts.
- This constriction results from the decreasing secretion of hormones, primarily **progesterone**, by the degenerating corpus luteum.
- In addition to vascular changes, the hormone withdrawal results in the stoppage **of glandular secretion, a loss of interstitial fluid, and a marked shrinking of the endometrium.**

Ischemic Phase cont...

- This results in *venous stasis* and patchy ischemic necrosis (death) in the superficial tissues.
- Eventually, rupture of damaged vessel walls follows and blood seeps into the surrounding connective tissue.
- As small pieces of the endometrium detach and pass into the uterine cavity, the torn ends of the arteries bleed into the uterine cavity, resulting in a loss of **20 to 80 mL of blood.**

Ischemic Phase cont...

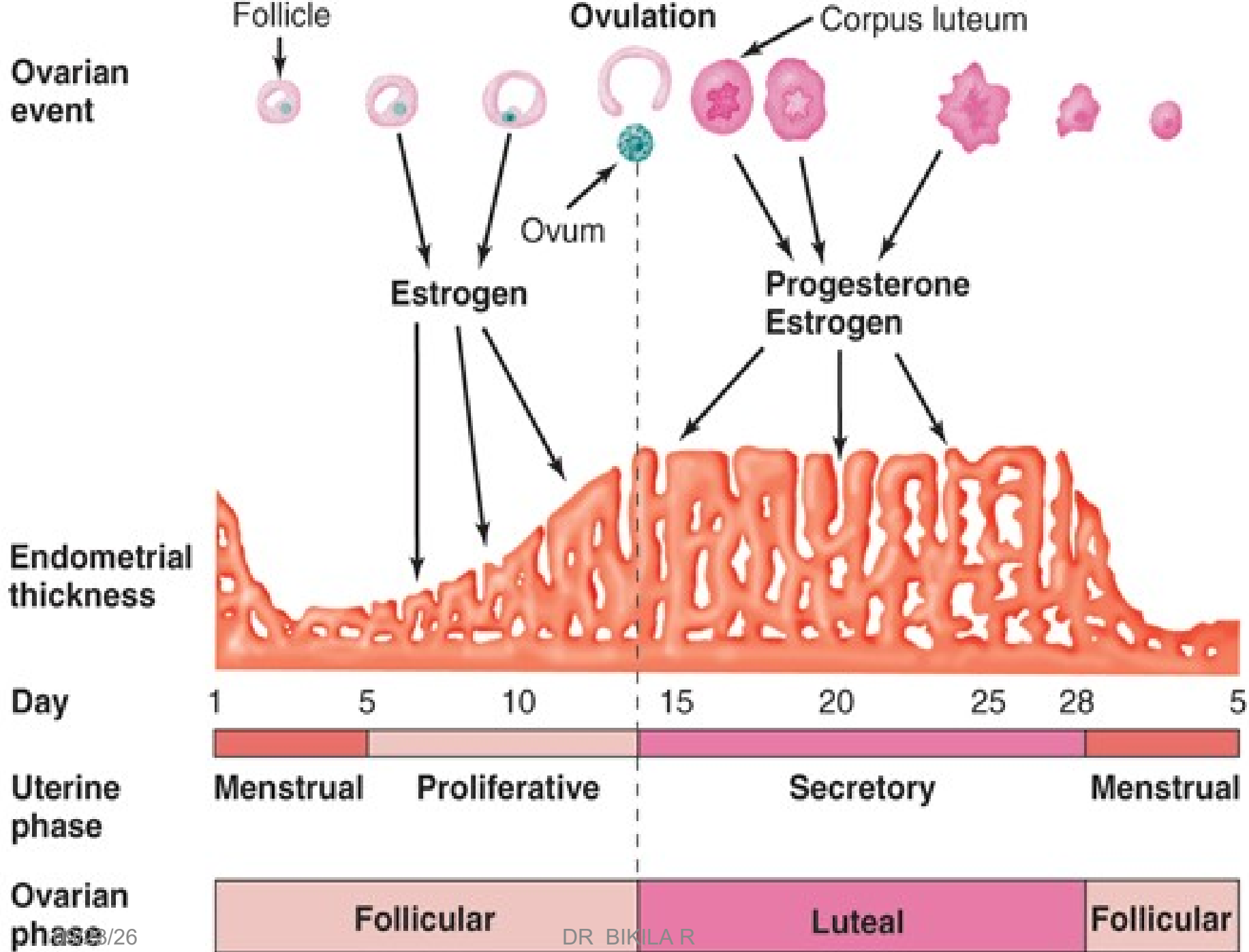
- Eventually, over 3 to 5 days, the entire compact layer and most of the spongy layer of the endometrium are discarded in the *menses*.
- Remnants of the **spongy and basal layers** remain to undergo regeneration during the subsequent proliferative phase of the endometrium.

If fertilization occurs:

- Cleavage of the zygote and blastogenesis (formation of blastocyst) occur.
- The blastocyst begins to implant in the endometrium on approximately the sixth day of the luteal phase (day 20 of a 28-day cycle).
- Degeneration of the **corpus luteum** is prevented by **human chorionic gonadotropin (hCG)**.
- The **luteal phase continues and menstruation does not occur.**

If fertilization occurs cont...

- **Luteal cells** continue to secrete **progesterone** for **11 to 12 week** of embryonic development;
- Thereafter , they regress slowly as secretion of **progesterone** by the **trophoblastic component of the placenta** becomes adequate for maintenance of pregnancy.
- Removal of the **corpus luteum** of pregnancy before the **fourth month** usually leads to **abortion**.



Transportation of Gametes

Oocyte transport

- During ovulation, the fimbriated end of the uterine tube becomes closely applied to the ovary.
- The fingerlike processes of the tube, *fimbriae*, move back and forth over the ovary.
- The sweeping action of the fimbriae and fluid currents produced by the **cilia** of the mucosal cells of the fimbriae "sweep" the secondary oocyte into the infundibulum of uterine tube.

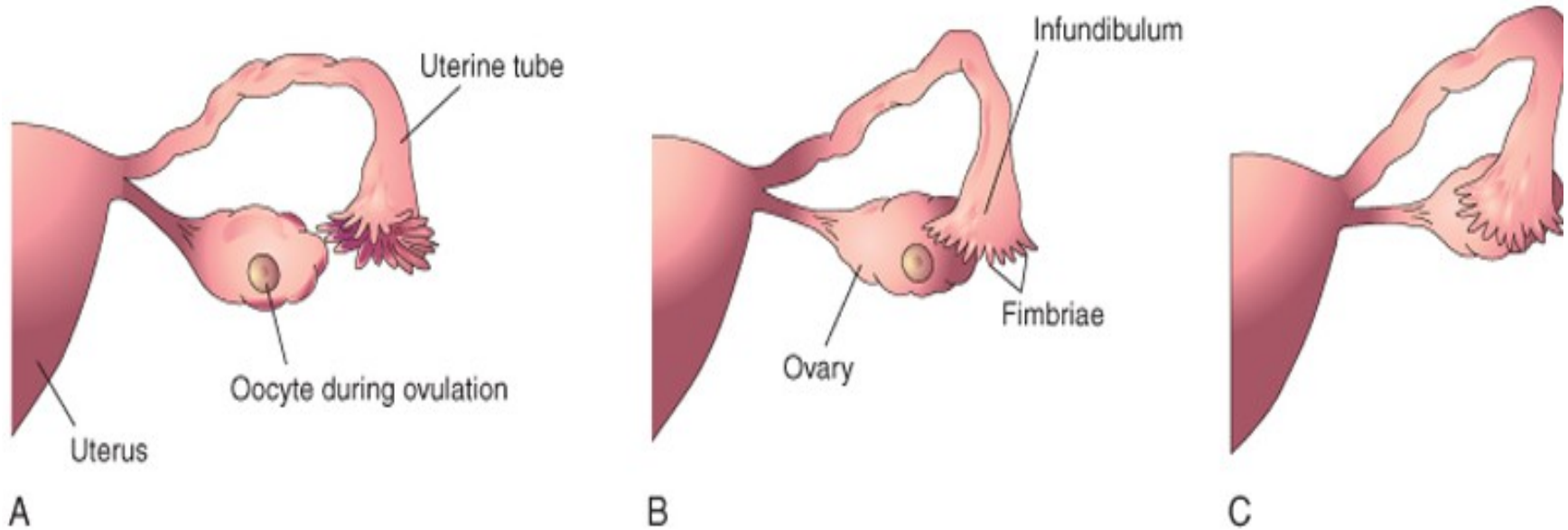


Figure: Illustrations of the movement of the uterine tube that occurs during ovulation. Note that the infundibulum of the tube becomes closely applied to the ovary. Its fingerlike fimbriae move back and forth over the ovary and "sweep" the secondary oocyte into the infundibulum as soon as it is expelled from the ovary during ovulation.

Sperm Transport

- From epididymis, the sperms are transported to the urethra by peristaltic contractions of the thick muscular coat of the ductus deferens.
- The accessory sex glands-***seminal glands (vesicles), prostate, and bulbourethral glands***-produce secretions that are added to the sperm-containing fluid in the urethra.

Sperm Transport cont...

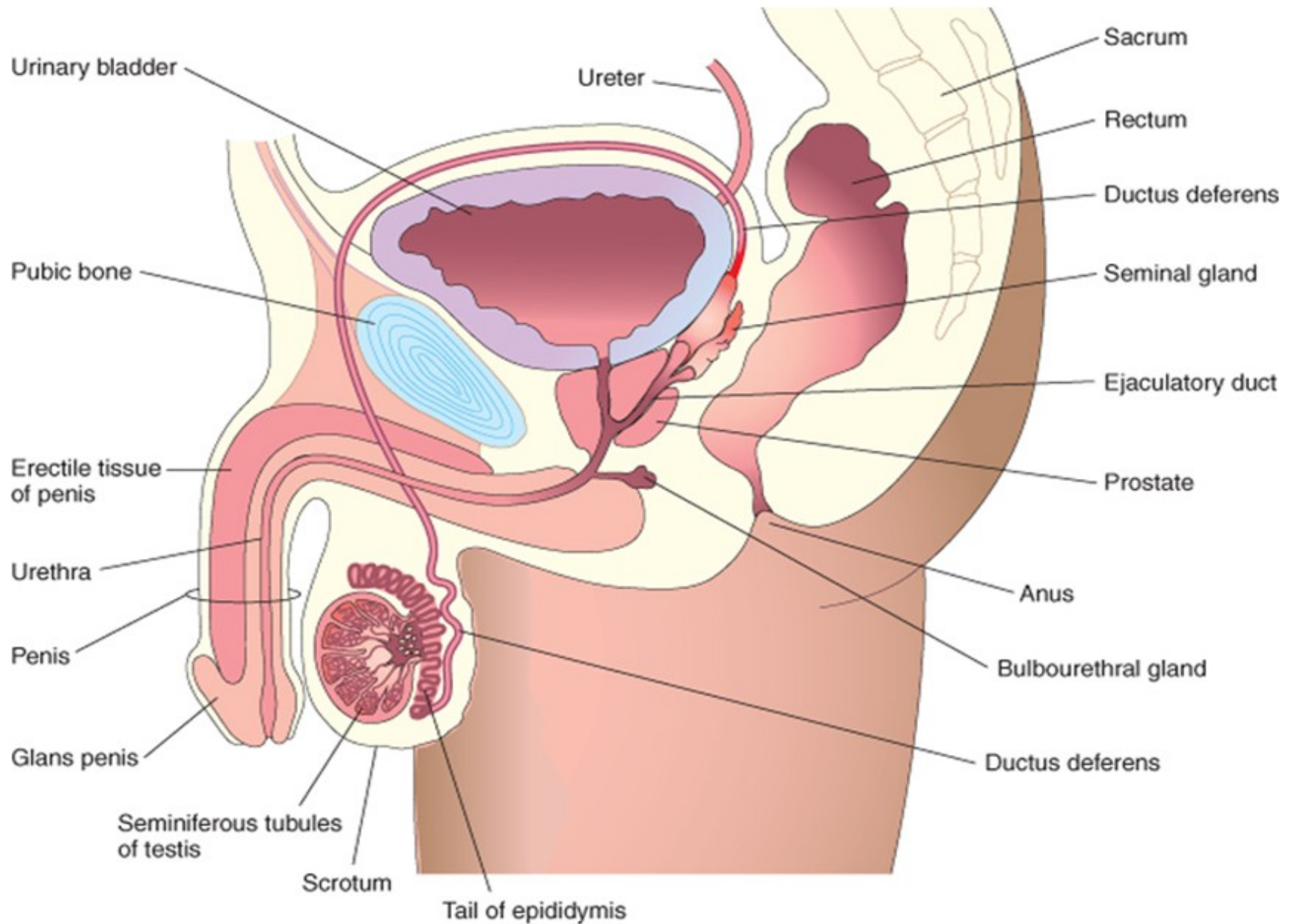
- From 200 to 600 million sperms are deposited around the **external os of the uterus** and in the **fornix of the vagina** during sexual intercourse.
- The sperms pass slowly through the cervical canal by movements of their tails.
- The enzyme **vesiculase**, produced by the seminal glands, **coagulates some of the semen (ejaculate)** and forms a **vaginal plug** that **may prevent the backflow of semen into the vagina..**

Sperm Transport cont...

- Passage of sperms through the uterus and uterine tubes results mainly from muscular contractions of the walls of these organs.
- *Prostaglandins* in the semen are thought to stimulate uterine motility at the time of intercourse and assist in the movement of sperms to the site of fertilization in the ampulla of the tube.
- *Fructose*, secreted by the seminal glands, is an energy source for the sperms in the semen.

The reflex ejaculation of semen may be divided into two phases:

- ***Emission:*** Semen is delivered to the prostatic part of the urethra through the ejaculatory ducts
- ***Ejaculation:*** Semen is expelled from the urethra through the external urethral orifice.



Maturation of sperms

- Freshly ejaculated sperms are unable to fertilize oocytes.
- *Sperms must undergo a period of conditioning-capacitation*-lasting approximately 7 hours.
- During this period, **a glycoprotein coat and seminal proteins** are removed from the surface of the sperm's acrosome.

Maturation of sperms cont...

- Sperms are usually capacitated in the uterus or uterine tubes by substances secreted by these parts of the female genital tract.
- During *in vitro fertilization*, capacitation is induced by incubating the sperms in a defined medium for several hours.
- Completion of capacitation permits the acrosome reaction to occur.

Viability of gametes

- Studies on early stages of development indicate that **human oocytes** are usually fertilized within 12 hours after ovulation.
- If oocyte cannot be fertilized within 24 hours, it will degenerate.
- Most **human sperms** probably do not survive for more than 48 to 72 hours in the female genital tract.
- Sperms and oocytes can be stored frozen for many years to be used in assisted reproduction

Fertilization

- Is the sequence of events that begins with contact between a **spermatozoa** and a **secondary oocyte** and
- Ends with the **fusion of the nuclei of the sperm and ovum** and the intermingling of maternal and paternal chromosomes at the **metaphase** of first mitotic division of the **zygote**.

Fertilization cont...

- Fertilization occurs in **ampulla** of the fallopian tube(widest and longest part)
- If the oocyte is not fertilized in the ampulla, it slowly passes along the tube to the uterus, where it degenerates and is resorbed
- Although fertilization may occur in other parts of the tube, *it does not occur in the uterus.*
- The fertilization process takes approximately 24 hours.

Fertilization cont.....

- The oocyte is viable for 12 to 24 hours
- Sperm is viable 24 to 72 hours
- **For fertilization to occur**, coitus must occur no more than:
 - Three days before ovulation
 - 24 hours after ovulation

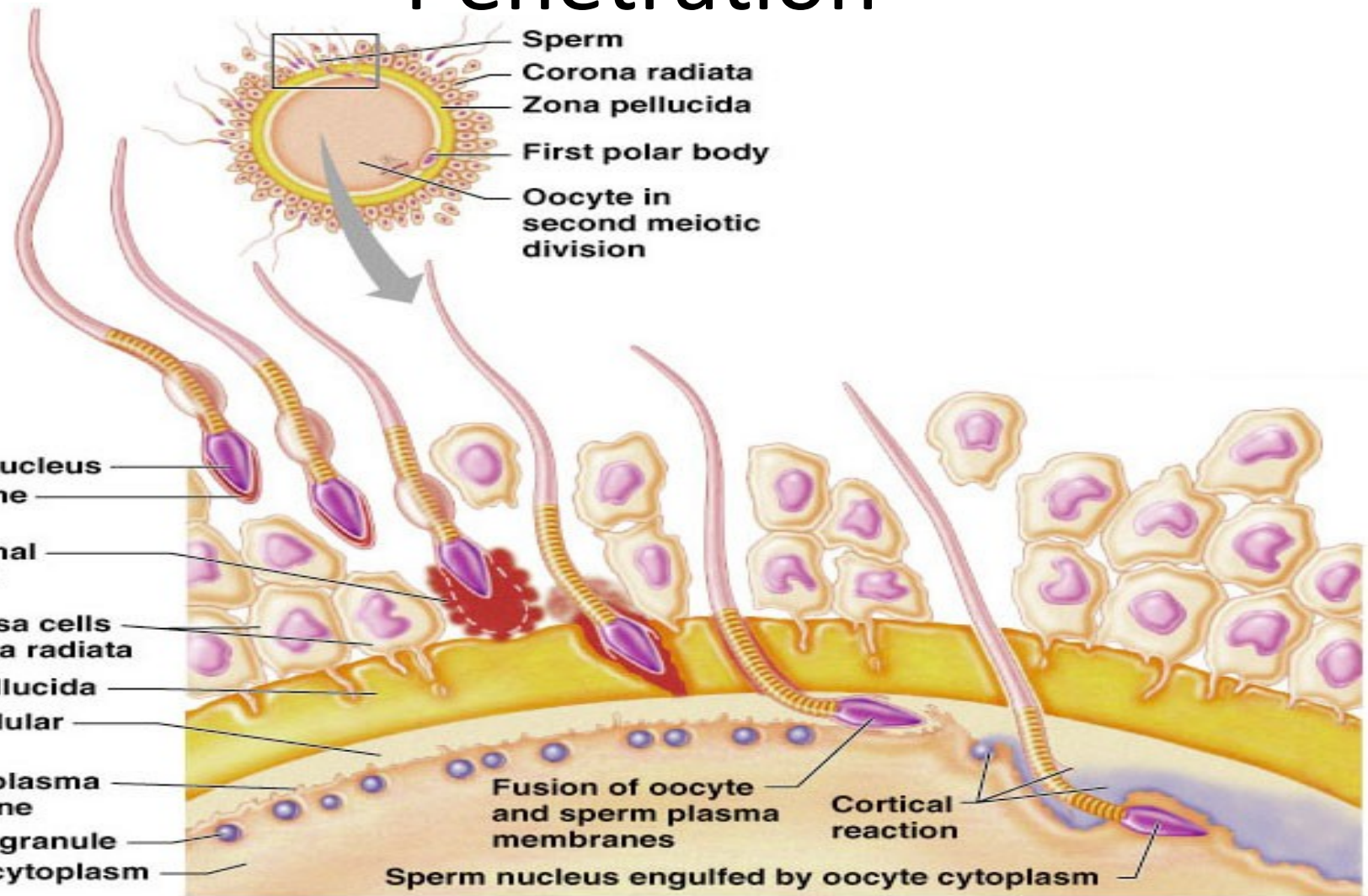
Fates of ejaculated sperm

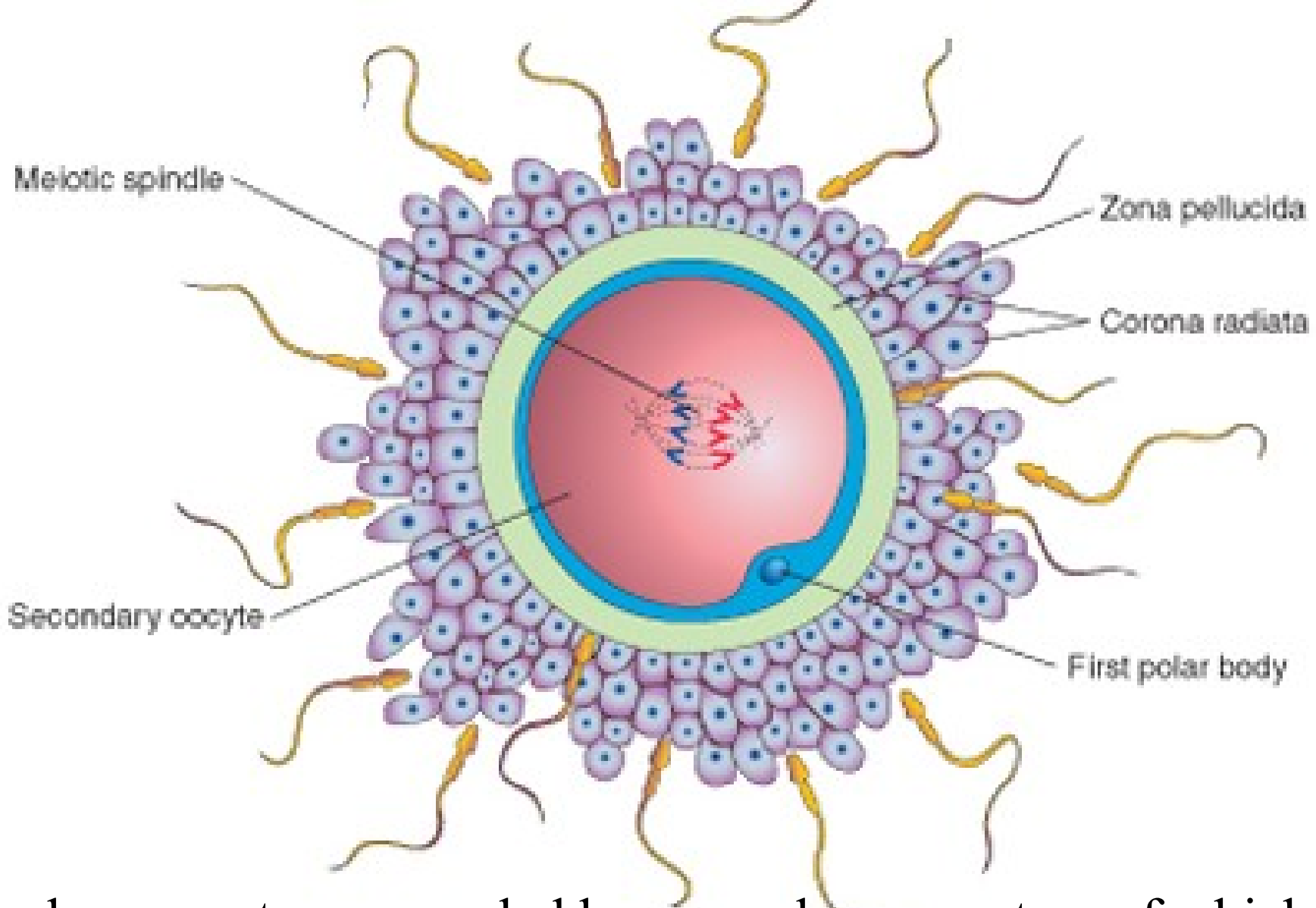
- Leak out of the vagina immediately after deposition
- Destroyed by the acidic vaginal environment
- Fail to make it through the cervix
- Dispersed in the uterine cavity or destroyed by phagocytic leukocytes
- Reach the uterine tubes
- Sperm must undergo capacitation before they can penetrate the oocyte

Steps of fertilization

1. The sperm passes through the **corona radiata**.
 - Enzyme hyaluronidase released from acrosome dissociates follicular cells.
2. The sperm penetrate the **zona pellucida (acrosomal reaction)**.
 - Enzymes esterases, acrosin and neuraminidase cause lyses of zona pellucida.
3. The sperm head attaches to the surface of **secondary oocyte** and fusion of cell membranes.
4. The *secondary oocyte completes the second meiotic division*, resulted in mature ovum and second polar body.
5. Male and female pronuclei fuse, forming a new cell called **zygote**.

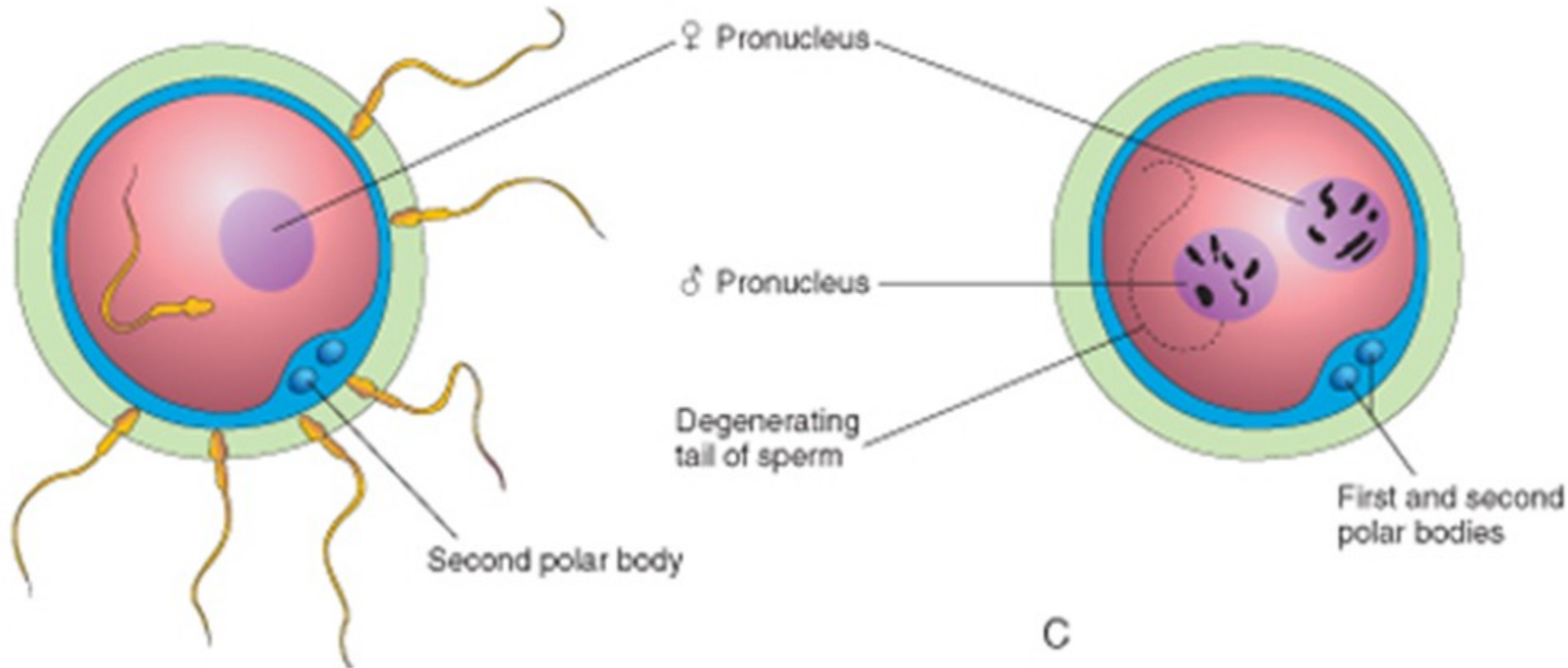
Acrosomal Reaction and Sperm Penetration





A

Secondary oocyte surrounded by several sperms, two of which have penetrated the corona radiata.



B, The corona radiata has disappeared, a sperm has entered the oocyte, and the second meiotic division has occurred, forming a mature oocyte. The nucleus of the oocyte is now the female pronucleus. **C**, The sperm head has enlarged to form the male pronucleus. This cell, an ootid, contains the male and female pronuclei.

Blocks to Polyspermy

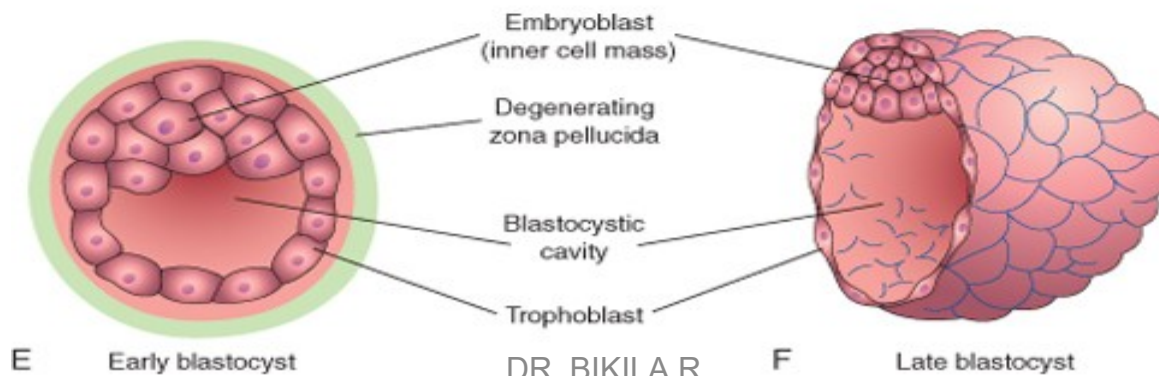
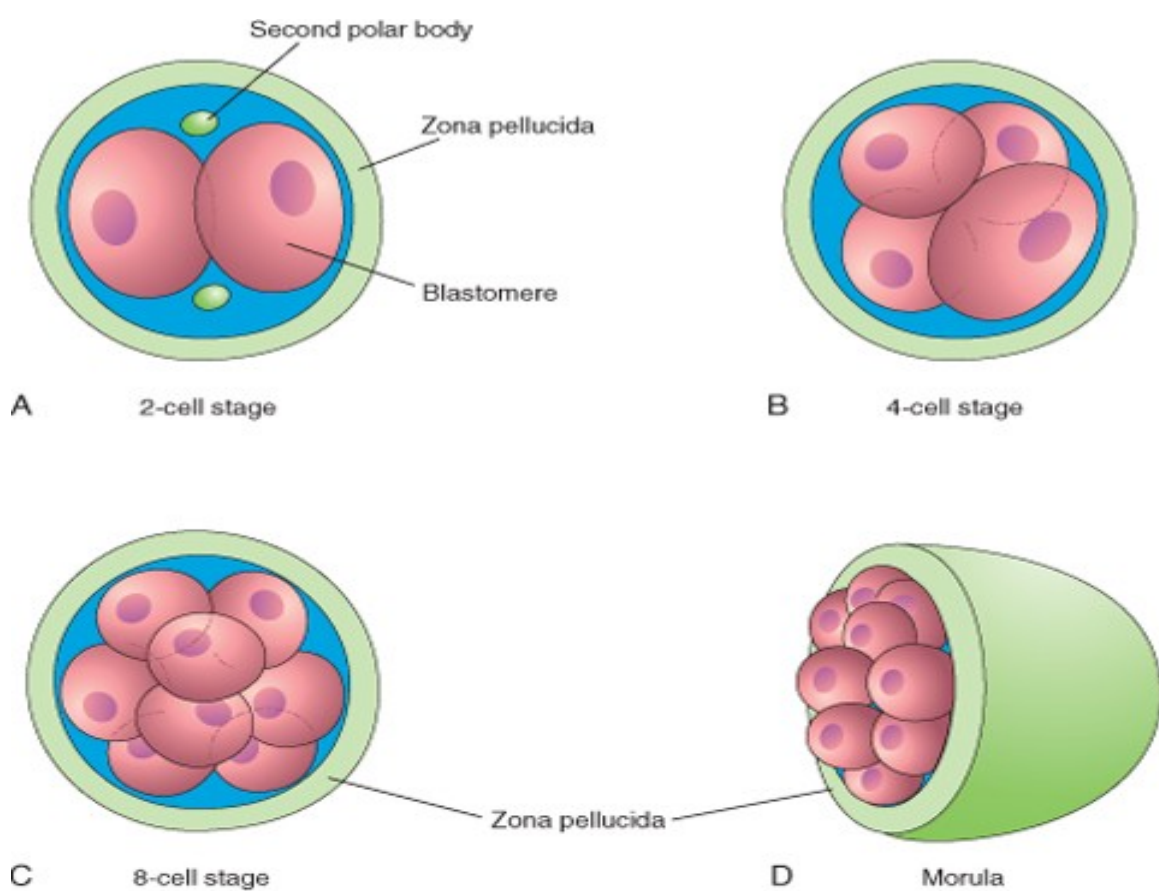
- Only one sperm is allowed to penetrate the oocyte
- **Two mechanisms ensure monospermy**
 - 1. Fast block to polyspermy** – membrane depolarization prevents sperm from fusing with the oocyte membrane
 - 2. Slow block to polyspermy** – zonal inhibiting proteins (ZIPs):
 - Destroy sperm receptors
 - Cause sperm already bound to receptors to detach

Results of fertilization

- **Stimulates the penetrated oocyte to complete the second meiotic division.**
- **Restores the normal diploid number of chromosomes (46) in the zygote.**
- **Results in variation of the human species through mingling of maternal and paternal chromosomes.**
- **Determines chromosomal sex of the embryo.**
- **Causes metabolic activation of the ootid and initiates cleavage (cell division) of the zygote**

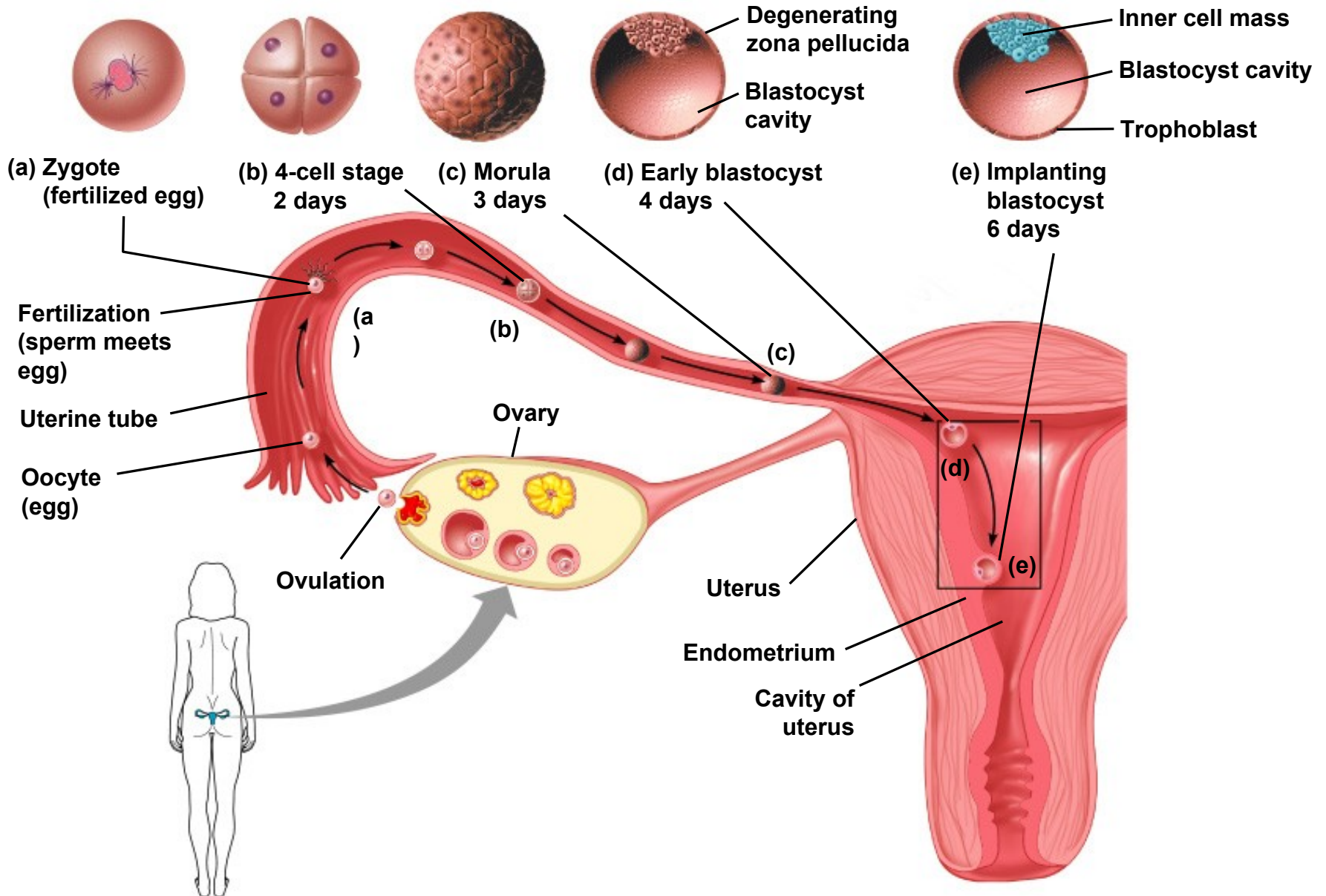
Cleavage of the zygote

- Cleavage consists of repeated **mitotic division** of the zygote, resulting in a rapid increase in number of cells.
- It **occurs in uterine tube** while the zygote is still contained in the **zona pellucida**.
- It begins about 30 hrs after fertilization.
- The resulting daughter cells are termed as *blastomeres*.
- 12 to 16 blastomeres form *morula* at about 3 days after fertilization and enter the uterine cavity.
- During this period, blastomeres are surrounded by the zona pellucida, which disappears at the end of the fourth day.



Cleavage of the zygote cont...

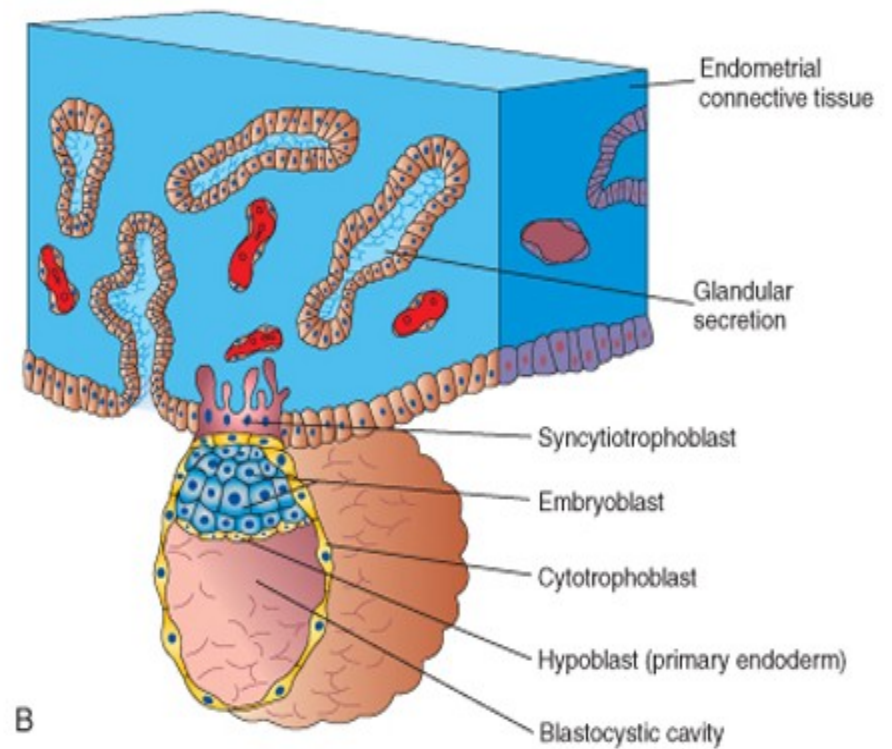
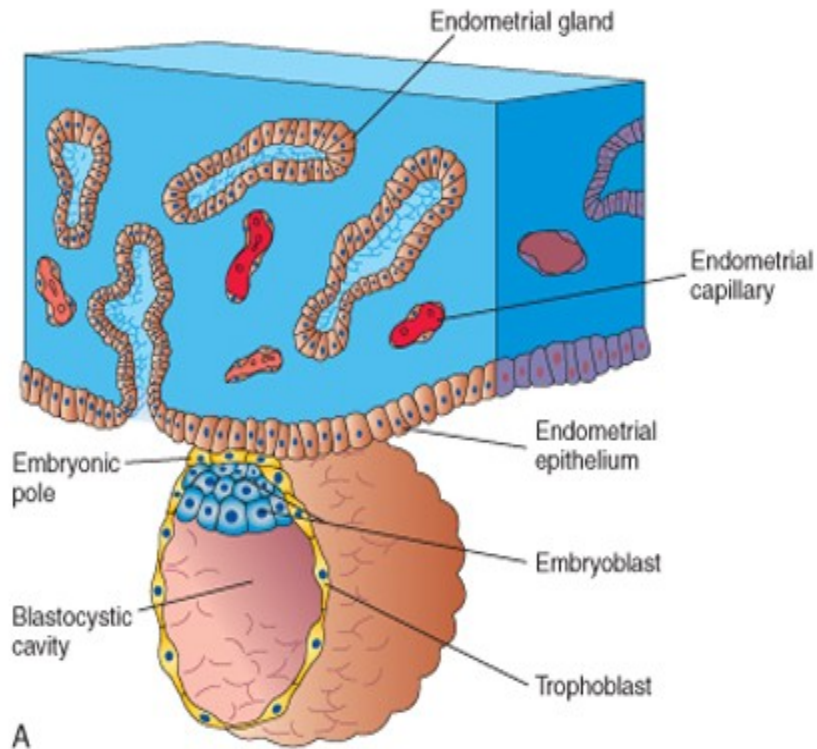
- About 4 days after fertilization, spaces appear between the central cells of the **morula** to form **blastocystic cavity**.
- Fluids pass into these cavities from uterine cavity.
- As fluid increases ,it separates the cells into 2 parts
 - Outer single cell layer, *Trophoblast* (take part in placenta formation)
 - Inner cell mass, *embryoblast* (becomes the embryonic disc)
- This fluid filled cell mass is called **blastocyst**.



Implantation

- Begins six to seven days after ovulation when the **trophoblasts** adhere to a properly prepared endometrium
- The **trophoblasts** then proliferate and form two distinct layers
 - **Cytotrophoblast** – cells of the inner layer that retain their cell boundaries
 - **Syncytiotrophoblast** – cells in the outer layer that lose their plasma membranes and invade the endometrium, are **secretory cells(hCG)**

Implantation cont...



Implantation cont.....

- Viability of the corpus luteum is maintained by **human chorionic gonadotropin (hCG)** secreted by the **trophoblasts**
- hCG prompts the corpus luteum to continue to secrete progesterone and estrogen
- Between the second and third month, the **placenta**:
 - Assumes the role of progesterone and estrogen production
 - Is providing nutrients and removing wastes

SUMMARY OF THE FIRST WEEK

- After ovulation, the oocyte travel into the ampulla where it may be fertilized.
- Ejaculation of semen in the vagina allows several hundred sperms pass through the uterus and enter the uterine tubes.
- When an *oocyte is contacted by a sperm*, it completes the *second meiotic division(mature oocyte and a second polar body are formed)*
- Fertilization is complete when the male and female pronuclei unite

SUMMARY OF THE FIRST WK cont...

- Zygote undergoes cleavage to form blastomeres.
- Approximately 3 days after fertilization, a ball of 12 or more blastomeres-a morula-enters the uterus
- The morula converted into a blastocyst.
- Four to *5 days after fertilization*, the *zona pellucida is shed*.
- By the end of the first week, the blastocyst is superficially implanted in the endometrium